

Online Decision-Making with Functional Covariates

Key words:

1. Introduction

2. Problem formulation

2.1. Notations

Let $\mathcal{H}$ be a separable Hilbert space with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and norm $\|\cdot\|_{\mathcal{H}}$. Specifically, let $\mathcal{L}^2[0, 1]$ be the space of square-integrable functions on $[0, 1]$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$, $f, g \in \mathcal{L}^2[0, 1]$, and norm $\|f\| = \langle f, f \rangle^{1/2}$. An operator on a Hilbert space $\mathcal{H}$ is a linear map $\mathbf{T} : \mathcal{H} \rightarrow \mathcal{H}$. We denote linear operators on $\mathcal{H}$ by bold uppercase letters such as $\mathbf{T}, \mathbf{A}, \mathbf{B}$. Specifically, let $\mathbf{1}$ denote the identity operator on $\mathcal{L}^2[0, 1]$. For two functions f, h in $\mathcal{L}^2[0, 1]$, let $f \otimes h$ denote the rank-one operator defined by $(f \otimes h)g = \langle f, g \rangle h$. An operator $\mathbf{T}$ is called *self-adjoint* if $\langle f, \mathbf{T}g \rangle = \langle \mathbf{T}f, g \rangle$ for all $f, g \in \mathcal{H}$. A self-adjoint operator $\mathbf{T}$ is said to be *positive semidefinite* (PSD) if $\langle f, \mathbf{T}f \rangle \geq 0$ for all $f \in \mathcal{H}$, in which case we write $\mathbf{T} \succeq 0$. Given two self-adjoint operators $\mathbf{A}$ and $\mathbf{B}$, we say that $\mathbf{A}$ *dominates* $\mathbf{B}$, written $\mathbf{A} \succeq \mathbf{B}$, if $\langle f, \mathbf{A}f \rangle \geq \langle f, \mathbf{B}f \rangle$ for all $f \in \mathcal{H}$, equivalently $\mathbf{A} - \mathbf{B} \succeq 0$. Let $R : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be a symmetric, square-integrable, nonnegative-definite function. We define the associated *integral operator* $\mathbf{L}_R : \mathcal{L}^2[0, 1] \rightarrow \mathcal{L}^2[0, 1]$ by

$$(\mathbf{L}_R f)(s) = \int_0^1 R(s, t) f(t) dt, \quad \forall f \in \mathcal{L}^2[0, 1], s \in [0, 1].$$

For convenience, we will write the composition $\mathbf{L}_A \mathbf{L}_B$ of two integral operators $\mathbf{L}_A$ and $\mathbf{L}_B$ as $\mathbf{L}_{AB}$. For two sequences $\{a_T\}, \{b_T\}$, we write $a_T \lesssim b_T$ or $a_T = O(b_T)$ (and equivalently $b_T \gtrsim a_T$ or $b_T = \Omega(a_T)$) if $\limsup_{T \rightarrow \infty} |a_T|/|b_T| < +\infty$.

2.2. The decision-making model

Consider the following sequential decision making model with two arms. At each time step $t \in [T]$, nature draws a covariate $X_t \in \mathcal{X} \subseteq \mathcal{L}^2[0, 1]$ from a centered distribution $\mathcal{F}$ on $\mathcal{L}^2[0, 1]$. The decision maker then chooses an action $a_t \in [m]$ and observes the corresponding reward:

$$y_t = \langle \beta^{(a_t)}, X_t \rangle + \varepsilon_t, \tag{1}$$

where ε_t is a σ^2 -subgaussian noise variable and $\beta^{(a)}, a \in [m]$ are functional coefficients in $\mathcal{L}^2[0, 1]$. We assume that the covariate sequence $X_1, X_2, \dots, X_T$ and the noise sequence $\varepsilon_1, \dots, \varepsilon_T$ are all independent with each other.

Let $\mathcal{G}_t = \{(X_s, a_s, y_s) : 1 \leq s \leq t\}$ denote the available history up to time t . A (possibly randomized) policy $\pi = \{\pi_t\}_{t=1}^T$ is a sequence of decision rules where each decision rule π_t is a map from $\mathcal{G}_{t-1} \times \mathcal{X}$ to a probability distribution on $\{0, 1\}$: at round t , given $(\mathcal{G}_{t-1}, X_t)$, the policy π_t selects an action a_t by sampling from $\pi_t(\mathcal{G}_{t-1}, X_t)$. Deterministic policies are the special case where π_t assigns all probability mass to a single action. For each covariate X_t , the optimal action is

$$a_t^*(X_t) = \arg \max_{a \in [m]} \langle \beta^{(a)}, X_t \rangle,$$

which maximizes the expected reward given X_t . The regret of a policy π up to horizon T is

$$\text{Reg}_\pi(T) = \mathbb{E} \left[\sum_{t=1}^T (\langle \beta^{(a_t^*(X_t))}, X_t \rangle - \langle \beta^{(a_t)}, X_t \rangle) \right] \quad (2)$$

where the expectation is taken over the randomness of π . That is, the regret $\text{Reg}_\pi(T)$ is the expected cumulative loss in reward incurred by following π instead of always playing the optimal action. The decision-maker's goal is to design a policy π to minimize $\text{Reg}_\pi(T)$.

2.3. The reproducing kernel Hilbert space (RKHS) framework

In this paper, we adopt the reproducing kernel Hilbert space (RKHS) framework from [Cai and Yuan \(2012\)](#), which assumes that $\beta^{(a)}, a \in [m]$ reside in some RKHS. Formally, let $K : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be a real, symmetric, square-integrable, and nonnegative definite function. Every such kernel K defines a Hilbert space of functions, denoted by $\mathcal{H}_K$, with inner product denoted by $\langle \cdot, \cdot \rangle_{\mathcal{H}_K}$. The inner product is chosen so that the *reproducing property* holds:

$$f(s) = \langle f, K(s, \cdot) \rangle_{\mathcal{H}_K}, \quad \forall f \in \mathcal{H}_K, s \in [0, 1].$$

Throughout this paper, we impose the following assumption:

ASSUMPTION 1. *The functional coefficients $\beta^{(a)}$ belong to an RKHS $\mathcal{H}_K$ associated with some kernel K for all $a \in [m]$. Without loss of generality, we assume $\|(\mathbf{L}_{K^{1/2}})^{-1} \beta^{(a)}\|_2 \leq 1$ for all $a \in [m]$.*

At a high level, the kernel K characterizes the complexity of estimating coefficients $\beta^{(a)}, a \in [m]$. A smoother kernel restricts the RKHS to simpler functions, making the estimation problem easier, while a more flexible kernel increases the difficulty of estimation. Typical choices include the polynomial kernel $K(s, t) = (1 + st)^d$, whose RKHS consists of polynomials of degree at most d ; the Gaussian (RBF) kernel $K(s, t) = \exp\left(-\frac{(s-t)^2}{2\sigma^2}\right)$, whose RKHS contains very smooth (analytic)

functions; and Sobolev-type kernels, which generate Sobolev spaces of functions with a specified number of square-integrable derivatives. Other kernels, such as the Brownian or Matérn family, can be introduced in the same way.

Following [Cai and Yuan \(2012\)](#), we assume for now that the kernel K is known exactly a priori. In Section 4, we also discuss the problem where K is only known to belong to a given family of kernels.

2.4. Assumptions on covariate distribution $\mathcal{F}$

In this section, we introduce the key assumptions on the covariate distribution $\mathcal{F}$. We first assume that the functional covariates X is subgaussian.

ASSUMPTION 2 (Sub-Gaussianity). *We assume $X \sim \mathcal{F}$ is Σ -subgaussian, where Σ is a self-adjoint, positive semidefinite operator. Concretely, for every $f \in \mathcal{L}^2[0, 1]$, the scalar random variable $\langle X, f \rangle$ is $\langle f, \Sigma f \rangle$ -subgaussian.*

Assumption 2 guarantees that covariate X concentrates in a sub-Gaussian way with covariance proxy Σ . In the finite-dimensional contextual bandit setting, an analogous assumption is commonly imposed [need to cite something here]: the covariate $x \in \mathbb{R}^d$ is assumed to have a finite sub-Gaussian norm σ_X^2 . Such concentration properties allow for tight control of confidence intervals in prediction, which in turn leads to minimax-optimal regret bounds. In our functional setting, we extend this assumption by employing a positive semidefinite operator Σ as the covariance proxy, thereby capturing the tail behavior along different directions in the functional covariate space.

It has been established in [Cai and Yuan \(2012\)](#) that the difficulty of the reward prediction task depends jointly on the kernel K and on the covariate distribution $\mathcal{F}$. To formalize this interaction, we define the transformed covariate

$$\tilde{X}_t = \mathbf{L}_{K^{1/2}} X_t, \quad \forall t \in [T],$$

and denote its distribution by $\tilde{\mathcal{F}}$. It is straightforward to see that $\tilde{X} \sim \tilde{\mathcal{F}}$ is also subgaussian with proxy

$$\tilde{\Sigma} := \mathbf{L}_{K^{1/2}} \Sigma \mathbf{L}_{K^{1/2}}.$$

We next introduce an additional assumption that specifies how the “energy” of $\tilde{\Sigma}$ is distributed across its spectrum. By the spectral theorem, there exists a sequence of positive eigenvalues $\sigma_1 \geq \sigma_2 \geq \dots > 0$ and a set of orthonormalized eigenfunctions $\phi_1, \phi_2, \dots$ such that

$$\tilde{\Sigma} = \sum_{k=1}^{\infty} \sigma_k \phi_k \otimes \phi_k.$$

We have the following assumption on the decaying rates of s_k .

ASSUMPTION 3. *There exists $r > 1/2$ such that $\sigma_k = O(k^{-2r})$. Without loss of generality, we further assume that the largest eigenvalue $\sigma_1 \leq 1$.*

Assumption 3 requires the eigenvalues $\{\sigma_k\}$ of $\tilde{\Sigma}$ to decay at a polynomial rate k^{-2r} , and thus characterizes how the “energy” of the covariance proxy $\tilde{\Sigma}$ is distributed across its spectrum. A larger value of r indicates that most of the energy is concentrated in the first few eigenfunctions, whereas a smaller r implies that the energy is more spread out across many directions. This characterization plays a crucial role in functional analysis in the sense that it enables us to effectively reduce the infinite-dimensional covariate space to its most “significant” directions. In Section 3.3, we will show explicitly how the parameter r influences the minimax regret rate.

We further note that Assumption 3 is consistent with the standard decay condition in RKHS analysis, which is the weakest assumption of its kind, to the best of our knowledge. Concretely speaking, Assumption 3 only requires that the eigenvalues σ_k decay sufficiently fast. By contrast, the FPCA literature typically imposes two stronger requirements. First, the eigenvalues must satisfy $\sigma_k \asymp k^{-2r}$, i.e., they are required to decay at a fixed polynomial rate. Second, FPCA assumes a separation (or “no clustering”) condition, meaning that consecutive eigenvalues are sufficiently separated to ensure identifiability of the associated eigenfunctions. Both of these conditions are substantially stronger than Assumption 3.

As a concluded remark of this section, Assumption 2 and Assumption 3 are satisfied simultaneously for a wide range of distributions $\mathcal{F}$ (and the corresponding $\tilde{\mathcal{F}}$). In particular, when $X \sim \mathcal{F}$ is a Gaussian process with covariance kernel C , Assumption 2 is satisfied with $\Sigma = \mathbf{L}_C$, and $\tilde{\Sigma}$ is always decaying with some positive rate r . In this case, Assumption 3 coincides with the eigenvalue decay condition for $\tilde{\Sigma} = \mathbf{L}_{K^{1/2}CK^{1/2}}$ stated in Theorem 2 of Cai and Yuan (2012). Beyond the Gaussian process setting, many other distributions also satisfy Assumptions 2 and 3. For instance, Lipschitz transformations of Gaussian processes, as well as mixtures of Gaussian processes, remain sub-Gaussian, and therefore continue to fall within our framework.

3. The decision-making algorithm based on point-wise confidence intervals

In this section, we introduce the decision-making algorithms for the problem formulated in the previous section. We first present an action elimination framework based on effective point-wise confidence intervals. We then discuss how to construct such confidence intervals via kernel ridge regression estimators under additional spectral dominance assumptions. We further introduce the margin condition and demonstrate how it enables a faster regret rate. Finally, we establish that the regret rate achieved by our algorithms is minimax optimal.

3.1. An action elimination framework for online decision-making

We first introduce a general action elimination framework, the **RegCB.Elimination** algorithm. The framework is first proposed by Foster et al. (2018) and is later utilized in many different contextual bandit settings. In Algorithm 1 we present a simplified version adapted by Li et al. (2024), which utilizes an offline oracle that takes samples as inputs and output confidence intervals for each (covariate, action) pair.

Algorithm 1 The action elimination with offline oracle (**RegCB.Elimination**)

- 1: **Algorithm parameters:** Time horizon T , offline oracle $\mathcal{O}$ with failure probability $\delta = m^{-1}T^{-3}$.
 - 2: Initialize: $\hat{\beta}_1^{(a)} \leftarrow 0, \forall a \in \{0, 1\}; \Delta_1^{(a)}(X) \leftarrow \Delta_0$ from oracle $\mathcal{O}, \forall X \in \mathcal{X}, a \in \{0, 1\}$
 - 3: **for** each epoch $\tau = 1, 2, \dots$ until T time periods have elapsed **do**
 - 4: **for** t in the next $n_\tau = 2^\tau$ periods **do**
 - 5: Observe the functional covariate $X_t \sim \mathcal{F}$;
 - 6: Eliminate the inferior actions: $\mathcal{A}_t \leftarrow \{a : \langle X_t, \hat{\beta}_\tau^{(a)} \rangle + \Delta_\tau^{(a)}(X_t) \geq \max_{a \in [m]} \langle X_t, \hat{\beta}_\tau^{(a)} \rangle - \Delta_\tau^{(a)}(X_t)\}$
 - 7: Choose an action uniformly at random: $a_t \leftarrow \text{Uniform}(\mathcal{A}_t)$;
 - 8: Observe the reward $y_t = \langle \beta^{(a_t)}, X_t \rangle + \varepsilon_t$;
 - 9: **end for**
 - 10: Update the estimations $\beta_{\tau+1}^{(a)}$ and $\Delta_{\tau+1}^{(a)}(\cdot)$ for $a \in [m]$ by feeding the samples $\{(X_i, a_i, y_i), i \in [2^{\tau-1} + 1, 2^\tau]\}$ into the oracle $\mathcal{O}$;
 - 11: **end for**
-

Algorithm 1 takes the time horizon T and an offline oracle $\mathcal{O}$ as inputs. The offline oracle $\mathcal{O}$ is a black-box procedure that takes a collection of samples as input and outputs an estimator $\hat{\beta}^{(a)}$ and point-wise confidence radius $\Delta^{(a)}(\cdot) : \mathcal{X} \rightarrow \mathbb{R}^+$ satisfying

$$\left| \langle X, \hat{\beta}^{(a)} \rangle - \langle X, \beta^{(a)} \rangle \right| \leq \Delta^{(a)}(X) \quad (3)$$

with probability at least $1 - \delta$ for all $X \in \mathcal{X}$ and $a \in [m]$. That is, the oracle $\mathcal{O}$ guarantees that the true reward $\langle X, \beta^{(a)} \rangle$ lies in the confidence interval $[\langle X, \hat{\beta}^{(a)} \rangle - \Delta^{(a)}(X), \langle X, \hat{\beta}^{(a)} \rangle + \Delta^{(a)}(X)]$ with high probability for all $X \in \mathcal{X}, a \in [m]$. Specifically, when there is no samples available, the oracle $\mathcal{O}$ outputs a Δ_0 such that

$$\left| \langle X, \beta^{(a)} \rangle \right| \leq \Delta_0 \quad (4)$$

with probability higher than $1 - \delta$ for all $a \in [m]$.

The algorithm then first divide the whole horizon into $\mathcal{O}(\log T)$ geometric epochs where each epoch has $2^\tau, \tau \in \mathbb{N}$ time periods. At the beginning of epoch τ , that is, at the end of last epoch $\tau - 1$, the algorithm calls the oracle $\mathcal{O}$ to update the estimator $\widehat{\beta}_\tau^{(a)}$ and point-wise confidence radius $\Delta_\tau^{(a)}(\cdot) : \mathcal{X} \rightarrow \mathbb{R}^+$ for each action $a \in \{0, 1\}$ by taking in the samples $(X_t, y_t, a_t), t \in [2^{\tau-1} + 1, 2^\tau]$ collected in epoch $\tau - 1$. And then, for each time period in epoch τ , upon observing the functional covariate X_t , the algorithm will decide whether to eliminate some actions for that covariate. We know that with high probability, the maximum reward is at least $\max_{a \in [m]} \langle X_t, \widehat{\beta}_\tau^{(a)} - \Delta_\tau^{(a)}(X_t) \rangle$. Thus we eliminate the actions whose upper confidence bounds cannot reach this value; i.e., we eliminate the actions that have no potential in being the best action given covariate X_t . After eliminating such actions from the whole action set $[m]$, the algorithm chooses an action from the rest of the actions uniformly at random.

Compared to other decision-making algorithms such as OFUL in [Abbasi-Yadkori et al. \(2011\)](#) and FALCON in [Simchi-Levi and Xu \(2022\)](#), the **RegCB.Elimination** framework offers several advantages. First of all, in the finite action setting, OFUL is suboptimal in covariate dimensions, while **RegCB.Elimination** achieves tight dependency to the covariate dimension. This superior dimension dependence is particularly important in the functional setting we consider, as the effective dimension of functional covariates will grow with sample size and cannot be neglected. Additionally, OFUL updates confidence intervals in a fully online manner which creates complex dependencies among samples. The **RegCB.Elimination** framework instead uses an offline oracle with i.i.d. samples within each epoch and thus is simpler for analysis. On the other hand, although FALCON also employs an offline oracle and achieves optimal regret in finite-action low-dimensional linear contextual bandits, it does not adapt to margin conditions since its offline oracle only calculates estimation error on the population level. The **RegCB.Elimination** algorithm uses a point-wise confidence interval oracle $\mathcal{O}$ that enables margin-adaptive exploration, yielding improved regret when favorable margin conditions hold (see Theorem 4).

Let τ_0 denote the total number of epochs executed when Algorithm 1 exits, and let $\mathcal{T}_\tau$ denote the set of time periods in epoch τ . We now present a theorem stating that the regret achieved by the **RegCB.Elimination** framework is actually tied to the width of the confidence intervals generated by the offline oracle $\mathcal{O}$.

THEOREM 1. *Suppose that Assumptions 1 to 3 hold. The regret of the **RegCB.Elimination** framework in Algorithm 1 satisfies*

$$\begin{aligned} \mathbb{E} [\text{Reg}_{\text{Alg. 1}}(T)] &\leq \mathcal{O} \left(\sum_{\tau=2}^{\tau_0} 2^\tau \sum_{a \in [m]} \mathbb{E} \left[(\Delta_\tau^{(a)}(X) + \Delta_\tau^{(a^*(X))}(X)) \right. \right. \\ &\quad \left. \left. \cdot \mathbb{1} \{ 0 < \left| \langle X, \beta^{(a^*(X))} - \beta^{(a)} \rangle \right| < 2\Delta_\tau^{(a)}(X) + 2\Delta_\tau^{(a^*(X))}(X) \} \right] \right) + \mathcal{O}(1). \end{aligned}$$

3.2. The confidence interval oracle $\mathcal{O}$ for functional regression

With Algorithm 1 and Theorem 1, we are in the place to provide an $\mathcal{O}$ such that it is valid, i.e., (4) and (3) holds.

Assuming Assumptions 1 to 3, it is straightforward to derive the following choice of Δ_0 by the sub-gaussianity of covariate X :

LEMMA 1. *Suppose that Assumptions 1 to 3 holds. We have (4) holds with $\Delta_0 \asymp b \log \delta$ with probability greater than $1 - \delta$ for all $a \in [m]$.*

In the remaining part of this section, we will introduce the roughness regularization estimators as well as the confidence interval associated with the estimator to serve as the oracle $\mathcal{O}$. The roughness regularization is widely used in nonparametric function estimation (see, e.g., Ramsay and Silverman (2005)) and is employed in RKHS framework in Yuan and Cai (2010), Cai and Yuan (2012). Formally, suppose that our oracle $\mathcal{O}$ takes samples of the form $\{(X_i, y_i, a_i), i \in [n]\}$. The oracle then estimates each of the functional coefficient $\beta^{(a)}$ respectively with the following roughness regularization method with a common regularized parameter λ_n :

$$\widehat{\beta}^{(a)} = \arg \min_{\beta \in \mathcal{H}_K} \sum_{i=1}^n \mathbb{1}\{a_i = a\} (y_i - \langle X_i, \beta \rangle)^2 + \lambda_n \|\beta\|_{\mathcal{H}_K}^2, \quad \forall a \in [m]. \quad (5)$$

For any $a \in [m]$, $\widehat{\beta}^{(a)}$ minimizes an objective with two terms. The first term, $\sum_{i=1}^n \mathbb{1}\{a_i = a\} (y_i - \langle X_i, \beta \rangle)^2$, is the total squared loss and measures the fit of β to the observations with action a . The second term is a regularization term, $\lambda_n \|\beta\|_{\mathcal{H}_K}^2$, that penalizes the squared RKHS norm of β . By penalizing the RKHS norm, this term penalizes energy in high-frequency modes; therefore, a smaller value of $\lambda_n \|\beta\|_{\mathcal{H}_K}^2$ corresponds to a smoother coefficient function β . Accordingly, the tuning parameter λ_n sets the balance between how closely the model fits the observed data and how smooth the coefficient function is. The optimal value of λ_n to achieve best trade-off is governed by the decaying rate r of the population covariance operator as we will show later in Theorem 2.

Unlike functional principal component analysis (FPCA), which truncates the expansion to a finite number of eigenfunctions, the roughness regularization in (5) smoothly downweights higher-order eigenfunctions rather than discarding them entirely. This avoids the need to choose a truncation level, which requires strong separation conditions on the eigenvalues and becomes highly sensitive when eigenvalues are closely clustered or tied. We refer readers to Cai and Hall (2006), Cai and Yuan (2012) for more detailed discussion. We also note that although (5) needs to minimize an objective on a infinite-dimensional space, the solution of (5) must be linear in X_i , and thus the final solution can be obtained by optimizing over the $\mathbb{R}^n$ space. We refer the readers to Yuan and Cai (2010) for the implementation details.

With the estimators $\hat{\beta}^{(a)}, a \in [m]$ in hands, we shall now consider constructing the suitable confidence radius $\Delta^{(a)}(X)$ for each pair $(X, a) \in \mathcal{X} \times [m]$ given the estimations $\hat{\beta}^{(a)}$. Before we introduce the specific forms of $\Delta(\cdot)$, we first introduce two additional assumptions regarding spectral dominance of the second-moment operator. For any $a \in [m]$, let

$$\mathcal{U}^{(a)} = \left\{ X \in \mathcal{X} : \langle X, \beta^{(a)} \rangle > \langle X, \beta^{(a')} \rangle, \forall a' \neq a \right\}$$

denote the set of covariates X such that $a^*(X) = a$, for all actions $a \in [m]$. Moreover let $M^{(a)}$ denote the second-moment kernel of the transformed covariate $\tilde{X} = \mathbf{L}_{K^{1/2}} X$ for $X \in \mathcal{U}^{(a)}, a \in [m]$. That is, let

$$M^{(a)}(s, u) = \mathbb{E}_{X \sim \mathcal{F}} \left[\tilde{X}(s) \tilde{X}(u) \mathbb{1}\{X \in \mathcal{U}^{(a)}\} \right].$$

We introduce the following two different assumptions on the dominance relationship between $\mathbf{L}_{M^{(a)}}, a \in [m]$ and the transformed covariance proxy $\tilde{\Sigma}$.

ASSUMPTION 4 (Weak dominance). *There exist constants $c_w > 0$ and $\mu > 0$ such that*

$$\tilde{\Sigma} + \mu \mathbf{1} \preceq c_w (\mathbf{L}_{M^{(a)}} + \mu \mathbf{1}), \quad \forall a \in [m].$$

The spectral dominance assumption in Assumption 4 extend the minimum-eigenvalue condition commonly imposed on the covariate covariance within optimal regions in finite-dimensional bandit models (Goldenshluger and Zeevi 2013, Bastani and Bayati 2020, Mao et al. 2025). In finite dimensional settings, this condition guarantees the local identifiability of arm-specific coefficients. In contrast, in the functional setting, the coefficients $\beta^{(a)}, a \in [m]$ cannot be fully identified from finitely many samples. The goal thus shifts from estimating entire coefficient functions to recovering the principal eigendirections of the global covariance operator that capture the dominant covariate variation. Assumption 4 ensures that the covariate distributions restricted to the optimal regions, $\mathcal{U}^{(a)}$, still span these leading directions of the overall distribution $\mathcal{F}$. To understand that from a concrete viewpoint, suppose that $\tilde{\Sigma}$ and $\mathbf{L}_{M^{(a)}}$ share same eigenfunctions for all $a \in [m]$. Then Assumption 4 is equivalent to requiring that the eigenvalues of $\tilde{\Sigma}$ larger than μ should be dominated by the corresponding eigenvalues of $\mathbf{L}_{M^{(a)}}$ (up to constant factor). Consequently, estimators trained on samples from the optimal regions remain informative for predicting rewards across all covariates.

Assumption 4 is automatically satisfied when the covariate process $X \sim \mathcal{F}$ is Gaussian. More generally, it holds whenever the covariate distribution allocates sufficient probability mass across the optimal regions $\mathcal{U}^{(a)}$, so that these regions collectively preserve the leading eigendirections of the population covariance $\tilde{\Sigma}$.

We now construct the confidence radius $\Delta^{(a)}(\cdot)$ with the estimators $\widehat{\beta}^{(a)}$ for $a \in [m]$ when Assumption 4 holds. We introduce the following notation. Recall that the oracle $\mathcal{O}$ takes samples $\{(X_i, y_i, a_i), i \in [n]\}$ as input. Let $\widetilde{X}_i = \mathbf{L}_{K^{1/2}} X_i$ denote the transformed covariates. Define the regularized empirical second-moment operator and the regularized optimal-region second-moment operator

$$\widehat{\mathbf{T}}_{\lambda_n}^{(a)} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{a_i = a\} \widetilde{X}_i \otimes \widetilde{X}_i + \lambda_n \mathbf{1}, \quad \mathbf{T}_{\lambda_n}^{*(a)} = \mathbf{L}_{M^{(a)}} + \lambda_n \mathbf{1}, \quad \forall a \in [m]. \quad (6)$$

The following lemma shows an important property of $\widehat{\mathbf{T}}_{\lambda_n}^{(a)}$, which tells it dominates the optimal-region operator $\mathbf{T}_{\lambda_n}^{*(a)}$ up to a constant factor.

LEMMA 2. *Suppose that Assumptions 1 to 3 hold and that Assumption 4 holds with $\mu = \lambda_n$. For any $a \in [m]$, $\delta \in (0, 1)$, if (3) holds, then there exists a constant $c > 0$ depending on c_w and m such that when $(c \log(T/\delta)/n)^{2r} \leq \lambda_n$,*

$$\frac{1}{2m} \mathbf{T}_{\lambda_n}^{*(a)} \preceq \widehat{\mathbf{T}}_{\lambda_n}^{(a)}$$

with probability greater than $1 - \delta/T$.

This property guarantees that the $\Delta^{(a)}(\cdot)$ in the following lemma is valid with high probability.

LEMMA 3. *Suppose that Assumptions 1 and 2 hold and that Assumption 4 holds with $\mu = \lambda_n$. For any $a \in [m]$, if*

$$\frac{1}{2m} \mathbf{T}_{\lambda_n}^{*(a)} \preceq \widehat{\mathbf{T}}_{\lambda_n}^{(a)},$$

then (3) holds with

$$\Delta^{(a)}(X) \asymp \sqrt{\lambda_n \log(T/\delta)} + \sqrt{\frac{\log(T/\delta)}{n} \langle \mathbf{L}_{K^{1/2}} X, (\widehat{\mathbf{T}}_{\lambda_n}^{(a)})^{-1} \mathbf{L}_{K^{1/2}} X \rangle}$$

with probability higher than $1 - \delta/T$ for all X and $a \in [m]$ when $(c \log(T/\delta)/n)^{2r} \leq \lambda_n$, where c is the constant from Lemma 2.

Lemma 3 shows the choice of $\Delta^{(a)}(\cdot)$ to serve as the oracle $\mathcal{O}$ such that (3) holds with high probability for any $a \in [m]$. We now compare the results in Lemma 3 with some similar results on prediction error in current functional regression literature. Cai and Hall (2006) studies a similar offline problem that aims to predict $\langle X, \beta^{(a)} \rangle$ for some given X and provides an upper bound on the prediction error via the FPCA method. However, their results only apply for smooth X ; i.e., X such that the corresponding coefficients under population eigenfunctions are also decaying. This assumption is stringent for the problem we are considering, as our covariates $X \in \mathcal{X}$ only have

decaying eigenvalues for the population covariance, and for single covariate X it can well be non-smooth. Many other literature, such as Cai and Yuan (2012), only provides the upper bound on population error rather than point-wise error, and cannot be adapted in our setting.

The proof of Lemma 3 is provided in Section EC.2. It establishes separate upper bounds for the prediction bias and variance. Given the empirical dominance in Lemma 2, the prediction bias is uniformly controlled over all covariates $X \in \mathcal{X}$ with high probability, which is crucial for attaining a minimax-optimal regret bound. The variance bound follows directly from standard concentration arguments on the noise terms (Cai and Yuan 2012).

3.3. Regret rate under oracle $\mathcal{O}$.

In this subsection, we shall present the regret of our Algorithm 1 using the oracle $\mathcal{O}$ we just introduced. With Theorem 1 and Lemma 3, it is straightforward to derive the following theorem on the cumulative expected regret.

THEOREM 2. *Suppose that Assumptions 1 to 3 hold. Choose $\lambda_n \asymp (\log(mT)/n)^{2r/(2r+1)}$ in (5) and suppose that Assumption 4 holds with $\mu = \lambda_T$. Using oracle $\mathcal{O}$ defined in eqs. (3) to (5), the regret of Algorithm 1 satisfies*

$$\mathbb{E} [\text{Reg}_{\text{Alg. 1}}(T)] \leq \tilde{\mathcal{O}} \left(T^{\frac{r+1}{2r+1}} \right).$$

Theorem 2 gives the upper bound on the expected cumulative regret under the optimal choice of $\lambda_n \asymp (\log(mT)/n)^{2r/(2r+1)}$, which balances the bias and variance terms in the confidence radius from Lemma 3. As we will show in the next theorem, the regret rate of $\tilde{\mathcal{O}}(T^{(r+1)/(2r+1)})$ cannot be further improved in the sense that for any policy, there always exists a “bad” problem instance such that the regret is at least of this order. Thus, the upper bound in Theorem 2 is minimax optimal.

THEOREM 3. *We have*

$$\inf_{\pi} \sup \text{Regret}_{\pi}(T) \geq \Omega(T^{\frac{r+1}{2r+1}}),$$

where the infimum is taken over all admissible policies π and the supremum is taken over all problem instances such that Assumptions 1 to 4 hold.

If we further restrict the problem class by imposing the following margin condition, a tighter regret bound holds. Let

$$\Delta(X) = \langle X, \beta^{(a^*(X))} \rangle - \max_{a' \neq a^*(X)} \langle X, \beta^{(a')} \rangle, \forall X \in \mathcal{X}$$

denote the margin value of covariate X ; i.e., the difference between the maximum reward and the second-maximum reward. The following assumption characterizes how $\Delta(X)$ is distributed around 0.

ASSUMPTION 5. *There exists $c_m > 0$ and $\alpha \in (0, 1 + 1/r)$ ¹ such that*

$$\mathbb{P}_{X \sim \mathcal{F}}[0 < \Delta(X) < h] \leq c_m h^\alpha.$$

The margin condition in Assumption 5 quantifies the probability that the suboptimality gap $\Delta(X)$ is small, with the exponent α governing the decay rate. A larger α implies that more probability mass is concentrated away from the decision boundary $\Delta(X) = 0$, making it easier to distinguish optimal actions and leading to a faster regret rate. The margin condition is widely adopted in finite-dimensional contextual bandits (see, e.g., Goldenshluger and Zeevi (2013), Bastani and Bayati (2020)), and Assumption 5 extends it naturally to the functional setting.

To utilize the margin condition in Assumption 5, we will also need the following dominance assumption which is a bit stronger than the dominance assumption in Assumption 4.

ASSUMPTION 6 (**Strong dominance**). *There exists a constant $c_s > 0$ such that*

$$\tilde{\Sigma} \preceq c_s \mathbf{L}_{M^{(a)}}, \quad \forall a \in [m].$$

Assumption 6 strengthens Assumption 4 by removing the regularization offset $\mu \mathbf{1}$. This stronger condition is needed to uniformly bound the variance term in the confidence radius of Lemma 3 over all $X \in \mathcal{X}$, which is required to exploit the margin condition.

THEOREM 4. *Suppose that Assumptions 1 to 3, 5 and 6 hold. Using oracle $\mathcal{O}$ defined in Equation (5), Equation (4) and Equation (3) and choosing $\lambda_n \asymp (\log(mT)/n)^{2r/(2r+1)}$ in (5) yields*

$$\mathbb{E} [\text{Reg}_{\text{Alg. 1}}(T)] \leq \tilde{\mathcal{O}} \left(T^{\frac{r+1-r\alpha}{2r+1}} \right).$$

Theorem 4 shows that with the stronger dominance assumption and the margin condition, a faster regret rate of $\tilde{\mathcal{O}}(T^{(r+1-r\alpha)/(2r+1)})$ can indeed be achieved. The proof of Theorem 4 is similar to the proof of Theorem 2 and is provided in Section EC.5. At a high level, the margin condition can help us govern the indicator $\mathbb{1}\{0 < |\langle X, \beta^{(a^*(X))} - \beta^{(a)} \rangle| < 2\Delta_\tau^{(a)}(X) + 2\Delta_\tau^{(a^*(X))}(X)\}$ in the regret decomposition in Theorem 1, and thus leads to a smaller regret rate.

¹ We restrict to $\alpha \in (0, 1 + 1/r)$ because for $\alpha \geq 1$, the regret in Theorem 4 reduces to $\tilde{\mathcal{O}}(\log T)$, matching the finite-dimensional rate and rendering the functional structure immaterial.

4. Model selection of kernel K

In the previous section, we assumed that the choice of kernel K and its associated RKHS $\mathcal{H}_K$, is known a priori. However, such exact knowledge of K is often hard to obtain in practice. For example, we may well assume that the functional coefficients $\beta^{(a)}, a \in [m]$ belong to some Sobolev space, but it is hard to know the smoothness level of such a Sobolev space. If we just choose a rough $\mathcal{H}_K$ that is excessively large to contain all $\beta^{(a)}$, the decision-making algorithm may suffer from degraded performance, as the decaying rate r in Assumption 3 can be too conservative for this choice of rough Sobolev space. However, if $\mathcal{H}_K$ is chosen to be too smooth to contain the true functional coefficients $\beta^{(a)}$, then we are taking the risk that the theoretical guarantees of the algorithm no longer hold. Therefore, when $\mathcal{H}_K$ is not known exactly, it is important to select a $\mathcal{H}_K$ as small as possible such that Assumption 1 holds. In this section, we will introduce the problem of selecting suitable RKHS while making online decisions and present a decision-making algorithm that can select optimal kernel in a kernel families in an online manner.

4.1. Problem formulation and assumptions

Formally, consider the following decision-making problem. Suppose that we have access to a family of l nested RKHS, denoted by $\mathcal{G} = \{\mathcal{H}_{K_1}, \mathcal{H}_{K_2}, \dots, \mathcal{H}_{K_l}\}$, where $\mathcal{H}_{K_1} \subseteq \mathcal{H}_{K_2} \subseteq \dots \subseteq \mathcal{H}_{K_l}$ and $\beta^{(a)} \in \mathcal{H}_{K_1}$ for all $a \in [m]$. We define the “optimal” RKHS in this family as

$$\mathcal{H}_{K_{s^*}} = \inf\{\mathcal{H} \in \mathcal{G} : \beta^{(a)} \in \mathcal{H}, \forall a \in [m]\}, \quad (7)$$

that is, $\mathcal{H}_{K_{s^*}}$ is the smallest space in $\mathcal{G}$ that contains all the functional coefficients. Ideally, when s^* is known, we can use Algorithm 1 for online decision making and achieve the optimal regret rate. However, here we shall consider a practical setting where we don't know the index of the optimal RKHS s^* . Our goal is still minimizing the regret defined in (2). A concrete example arises when each $\beta^{(a)}$ belongs to a Sobolev space of unknown smoothness. Specifically, we can construct $\mathcal{G}$ such that $\mathcal{H}_{K_s}, s \in [l]$ denote the Sobolev RKHS of order $s > 0$, consisting of functions whose s -th derivatives are square-integrable. While we do not know the smoothness of the RKHS that contains each $\beta^{(a)}$, we aim to design a decision-making algorithm that can adapt to different level of smoothness within the RKHS family $\mathcal{G}$.

We continue to adopt Assumption 2 from Section 2.4 which assumes that covariate X is subgaussian with covariance proxy Σ . For the model selection problem we consider, we shall assume that each kernel $K_s, s \in [k]$ is associated with a decaying rate r_s . Let

$$\tilde{\Sigma}_s = \mathbf{L}_{K_s^{1/2}} \Sigma \mathbf{L}_{K_s^{1/2}}$$

denote the transformed covariance proxy associated with kernel K and σ_k^s denote the k^{th} largest eigenvalue of $\tilde{\Sigma}_s$. We have the following assumption.

ASSUMPTION 7. *For each $s \in [l]$, there exists $r_s > 1/2$ such that $\sigma_k^s = \mathcal{O}(k^{-2r_s})$.*

Specifically, for the model selection problem we consider, we also introduce the following dominance assumption:

ASSUMPTION 8. *There exist $h > 0$ and a constant $c_o > 0$ such that for all $a \in [m]$,*

$$\Sigma \preceq c_o \mathbb{E}_{X \sim \mathcal{F}} \left[\mathbb{1}\{X \in \mathcal{U}_h^{(a)}\} X \otimes X \right],$$

where

$$\mathcal{U}_h^{(a)} = \left\{ X \in \mathcal{X} \mid \langle X, \beta^{(a)} \rangle - \max_{a' \neq a} \langle X, \beta^{(a')} \rangle > h \right\}.$$

[Some explanations about this assumption]

4.2. A decision-making algorithm for kernel selection

We now present our decision-making algorithm for model selection in Algorithm 2. Algorithm 2 adapt an action elimination framework similar to Algorithm 1, but is specifically taylorred for the need of model selection. Algorithm 2 executes in the following way. The algorithm will first random explore for $t_0 = \tilde{\mathcal{O}}(h^{-(2+1/r_1)})$ rounds. And then, the algorithm initializes the estimators and confidence radius based on samples collected in the random exploration phase *for each* model $s \in [l]$. That is, without knowing the true model, we still calculate a pseudo estimator and confidence radius for each of the model in the family $\mathcal{G}$.

Then, for the remaining $t - t_0$ rounds, the algorithm will calculate a “good” arm set for each model upon observing the covariate X_t using the psuedo estimators for each arm $a \in [m]$. Compared to Algorithm 1, where the true model is assumed to be known, Algorithm 2 can no longer choose an action by constructing a good action set for a specific model. Instead, Algorithm 2 constucts “pseudo” good arm set for each model $s \in [l]$. Similar to the elimination process in Algorithm 1, for each model $s \in [l]$, we eliminate the actions whose upper confidence bounds cannot reach the largest lower confidence bound, $\max_{a \in [m]} \langle X_t, \hat{\beta}_t^{(a)} \rangle - \Delta_{s,t}^{(a)}$. After specifying the good arm set $\mathcal{A}_t^s$ for each model $s \in [l]$, the algorithm then find the largest model index s in $[l]$, such that the intersection of the good arm set for model $\mathcal{H}_{K_1}$ to model $\mathcal{H}_{K_s}$, is not empty. And then the algorithm will choose an action from this non-empty intersection uniformly at random.

The idea of action elimination in Algorithm 2 is similar to that in Algorithm 1. However, Algorithm 2 is different in several ways. First of all, Algorithm 2 has an explicit random exploration phase while Algorithm 1 does not. This random exploration step is to exploit the dominance outside

Algorithm 2 The model selection algorithm for functional bandit

-
- 1: **Algorithm parameters:** Time horizon T , RKHS family $\mathcal{G} = \{\mathcal{H}_{K_s}, s \in [l]\}$, gap parameter h ; regularization parameters $\lambda_{s,t}, s \in [l], t \in [t_0 + 1, T]$;
 - 2: Initialize: failure probability $\delta = m^{-1}T^{-3}$;
 - 3: **for** t in first $t_0 = \tilde{\mathcal{O}}(h^{-(2+1/r_1)})$ time periods **do**
 - 4: Randomly explore: $a_t \leftarrow \text{Uniform}([m])$;
 - 5: **end for**
 - 6: Initialize the estimations $\hat{\beta}_{s,t_0+1}^{(a)}$ for each model $s \in [l]$ by feeding the samples $\{(X_i, a_i, y_i), i \in [t_0]\}$ into (5) with $\lambda_n = \lambda_{s,t_0}, K = K_s$;
 - 7: Initialize confidence radius:

$$\Delta_{s,t_0+1}^{(a)}(X) \asymp \sqrt{\lambda_{t_0}} \log(1/\delta) + \sqrt{\frac{\log(1/\delta)}{t_0}} \langle \mathbf{L}_{K_s^{1/2}} X, \left(\hat{\mathbf{T}}_{s,t_0}^{(a)} \right)^{-1} \mathbf{L}_{K_s^{1/2}} X \rangle, \forall X \in \mathcal{X}, a \in [m], s \in [l],$$

where

$$\hat{\mathbf{T}}_{s,t_0}^{(a)} = \frac{1}{t_0} \sum_{i=1}^{t_0} \mathbb{1}\{a_i = a\} \mathbf{L}_{K_s^{1/2}} X_i \otimes \mathbf{L}_{K_s^{1/2}} X_i + \lambda_{t_0} \mathbf{1}, \forall a \in [m], s \in [l];$$

- 8: **for** t in the remaining $t - t_0$ periods **do**
- 9: Observe the functional covariate $X_t \sim \mathcal{F}$;
- 10: **if** $\max_{a \in [m]} \langle X_t, \hat{\beta}_{1,t_0+1}^{(a)} \rangle \geq \max_{a' \in [m], a' \neq a} \langle X_t, \hat{\beta}_{1,t_0+1}^{(a)} \rangle + \delta$, **then**
- 11: Chooses the myopic optimal action: $a_t \leftarrow \arg \max_{a \in [m]} \langle X_t, \hat{\beta}_{1,0}^{(a)} \rangle$
- 12: **else**
- 13: Calculate the good arms set for each model: $\mathcal{A}_t^s \leftarrow \left\{ a : \langle X_t, \hat{\beta}_{s,t}^{(a)} \rangle + \Delta_{s,t}^{(a)} \geq \max_{a \in [m]} \langle X_t, \hat{\beta}_{s,t}^{(a)} \rangle - \Delta_{s,t}^{(a)} \right\}, \forall s \in [l]$.
- 14: Find the smallest non-empty action intersection set: $s_t \leftarrow \max\{s \in [l], \bigcap_{i=1}^s \mathcal{A}_t^i \neq \emptyset\}$;
- 15: Choose an action uniformly at random: $a_t \leftarrow \text{Uniform}(\mathcal{A}_t^{s_t})$;
- 16: **end if**
- 17: Observe the reward $y_t = \langle \beta^{(a_t)}, X_t \rangle + \varepsilon_t$;
- 18: Update the estimations $\hat{\beta}_{s,t+1}^{(a)}$ by feeding the samples $\{(X_i, a_i, y_i), i \in [t]\}$ into (5) with $\lambda_n = \lambda_t, K = K_s$ for all $a \in [m], s \in [l]$;
- 19: Update the confidence radius

$$\Delta_{s,t+1}^{(a)}(X) \asymp \sqrt{\lambda_{s,t}} \log(1/\delta) + \sqrt{\frac{\log(1/\delta)}{t}} \langle \mathbf{L}_{K_s^{1/2}} X, \left(\hat{\mathbf{T}}_{s,t}^{(a)} \right)^{-1} \mathbf{L}_{K_s^{1/2}} X \rangle, \forall X \in \mathcal{X}, a \in [m], s \in [l],$$

where

$$\hat{\mathbf{T}}_{s,t}^{(a)} = \frac{1}{t} \sum_{i=1}^t \mathbb{1}\{a_i = a\} \mathbf{L}_{K_s^{1/2}} X_i \otimes \mathbf{L}_{K_s^{1/2}} X_i + \lambda_{s,t} \mathbf{1}, \forall a \in [m], s \in [l];$$

- 20: **end for**
-

margin assumption in Assumption 8 and thus is not required in Algorithm 1 when Assumption 8 is not imposed. It also allows us to identify the optimal action for covariates X such that the associated margin $|\Delta(X)|$ is larger than δ in Algorithm 2. Second, Algorithm 2 uses a uniform confidence radius over all $X \in \mathcal{X}$, rather than the covariate-dependent radius in Algorithm 1. This simplification is needed because the algorithm compares good arm sets across models, and a covariate-dependent radius would make such comparisons intractable. Third, Algorithm 2 does not require the geometric batch structure of Algorithm 1. As we show in the proof of Lemma EC.2, the algorithm’s greedy step produces i.i.d. samples without explicit batching.

4.3. Regret analysis of decision-making algorithm with kernel selection

The following theorem shows the regret of Algorithm 2.

THEOREM 5. *Suppose that Assumptions 2, 3 and 8 hold. Using oracle $\mathcal{O}$ defined in (5), (4) and (3) and choosing $\lambda_{s,t} \asymp (\log(lmT)/t)^{2r_s/(2r_{s^*}+1)}$ for all $t \in [t_0 + 1, T], s \in [l]$, the regret of Algorithm 2 satisfies*

$$\mathbb{E}[\text{Regret}_{\text{Alg. 2}}(T)] = \tilde{\mathcal{O}}\left(T^{\frac{r_{s^*}+1}{2r_{s^*}+1}}\right).$$

Theorem 5 shows that by optimally balancing bias and variance for each model, Algorithm 2 achieves an expected regret of order $\tilde{\mathcal{O}}(T^{(r_{s^*}+1)/(2r_{s^*}+1)})$, which matches the regret rate achieved by Algorithm 1 when $\mathcal{H}_{K_{s^*}}$ is known in hindsight. Therefore, Algorithm 1 effectively chooses the right model, in the sense that the minimax regret rate is still achieved when the optimal model is not known.

To optimally balance the bias and variance for each of the model, the algorithm need to know r_s for each model $s \in [l]$. That is often possible to achieve, since r_s is the decaying rate of $\mathbf{L}_{K_s^{1/2}} \mathbf{\Sigma} \mathbf{L}_{K_s^{1/2}}$, and we know $\mathbf{L}_{K_s^{1/2}}$ for each $s \in [l]$. In contrast, many model selection bandit algorithms such as CORRAL (Agarwal et al. 2017) requires the regret rate of the optimal model to tune its hyperparameters which is not practical in our setting: without knowing K_{s^*} , it is almost impossible to know the decaying rate of the eigenvalues for the operator $\mathbf{L}_{K_{s^*}} \mathbf{\Sigma} \mathbf{L}_{K_{s^*}}$, neither the corresponding optimal regret rate $T^{(r_{s^*}+1)/(2r_{s^*}+1)}$. Our model selection algorithm is thus more practically applicable in functional setting.

REMARK 1. Remark on the margin condition can also improve the regret rate

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Omitted proofs

EC.1. Proof of Lemma 2

Fix any $a \in [m]$. Recall that $\widehat{\mathbf{T}}_{\lambda_n}^{(a)} = \widehat{\mathbf{T}}_n^{(a)} + \lambda_n \mathbf{1}$ where $\widehat{\mathbf{T}}_n^{(a)} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{a_i = a\} \widetilde{X}_i \otimes \widetilde{X}_i$. Let

$$\mathbf{T}_{\lambda_n}^{(a)} = \mathbb{E}[\mathbb{1}\{a_1 = a\} \widetilde{X}_1 \otimes \widetilde{X}_1] + \lambda_n \mathbf{1}$$

denote the population counterpart of $\widehat{\mathbf{T}}_{\lambda_n}^{(a)}$. We first show that $\widehat{\mathbf{T}}_{\lambda_n}^{(a)}$ concentrates around its population counterpart $\mathbf{T}_{\lambda_n}^{(a)}$ via Lemma EC.9. To this end, we show that the algorithm's sampling of arm a inherits sufficient mass from the region where a is optimal to verify the dominance condition (EC.35) in Lemma EC.9.

Define the good action set for each sample i as

$$\mathcal{A}_i = \{a : \langle X_i, \widehat{\beta}_\tau^{(a)} \rangle + \Delta_\tau^{(a)}(X_i) \geq \max_{a' \in [m]} \langle X_i, \widehat{\beta}_\tau^{(a')} \rangle - \Delta_\tau^{(a')}(X_i)\},$$

analogous to line 6 of Algorithm 1. By the algorithm design, when (3) holds, the optimal arm $a^*(X_i)$ is included in $\mathcal{A}_i$. To see this, note that (3) gives

$$\langle X, \widehat{\beta}_\tau^{(a^*)} \rangle + \Delta_\tau^{(a^*)}(X) \geq \langle X, \beta^{(a^*)} \rangle \geq \langle X, \beta^{(a)} \rangle \geq \langle X, \widehat{\beta}_\tau^{(a)} - \Delta_\tau^{(a)}(X) \rangle, \forall a \in [m], \forall X \in \mathcal{X}$$

where the first and the last inequality uses triangle inequality with (3) and the second inequality is given by the optimality of a^* . Since $a^*(X_i)$ remains in $\mathcal{A}_i$ and the algorithm selects a_i uniformly from at most m actions, action a is sampled with probability at least $1/m$ whenever it is the optimal arm. Consequently we have

$$\begin{aligned} \mathbf{T}^{(a)} &:= \mathbb{E}[\mathbb{1}\{a_i = a\} \widetilde{X}_i \otimes \widetilde{X}_i] \succeq \mathbb{E}[\mathbb{1}\{a_i = a, a^*(\widetilde{X}_i) = a\} \widetilde{X}_i \otimes \widetilde{X}_i] \\ &\succeq \frac{1}{m} \mathbb{E}[\mathbb{1}\{a^*(\widetilde{X}_i) = a\} \widetilde{X}_i \otimes \widetilde{X}_i] \\ &= \frac{1}{m} \mathbf{L}_{M^{(a)}}. \end{aligned} \tag{EC.1}$$

By Assumption 2, $\mathbb{1}\{a_i = a\} \widetilde{X}_i$ is $\widetilde{\Sigma}$ -sub-Gaussian. We verify the dominance condition (EC.35) with $\Sigma_0 = \mathbf{T}^{(a)}$, $\Sigma_{\text{proxy}} = \widetilde{\Sigma}$, and $\lambda = \lambda_n$. Recall from (6) that $\mathbf{T}_{\lambda_n}^{*(a)} = \mathbf{L}_{M^{(a)}} + \lambda_n \mathbf{1}$. By Assumption 4 with $\mu = \lambda_n$,

$$\widetilde{\Sigma} + \lambda_n \mathbf{1} \preceq c_w (\mathbf{L}_{M^{(a)}} + \lambda_n \mathbf{1}) = c_w \mathbf{T}_{\lambda_n}^{*(a)}. \tag{EC.2}$$

By (EC.1), $\mathbf{L}_{M^{(a)}} \preceq m \mathbf{T}^{(a)}$, so $\mathbf{T}_{\lambda_n}^{*(a)} = \mathbf{L}_{M^{(a)}} + \lambda_n \mathbf{1} \preceq m \mathbf{T}^{(a)} + \lambda_n \mathbf{1} \preceq m \mathbf{T}_{\lambda_n}^{(a)}$. Therefore,

$$\widetilde{\Sigma} + \lambda_n \mathbf{1} \preceq c_w m \mathbf{T}_{\lambda_n}^{(a)} = c_w m (\mathbf{T}^{(a)} + \lambda_n \mathbf{1}), \tag{EC.3}$$

and (EC.35) holds with $c_d = c_w m$. The effective dimension satisfies

$$d_{\text{eff}} = \text{Tr}(\tilde{\Sigma}(\mathbf{T}_{\lambda_n}^{(a)})^{-1}) + 1 \leq c_w m \text{Tr}(\tilde{\Sigma}(\tilde{\Sigma} + c_w m \lambda_n \mathbf{1})^{-1}) + 1 = \mathcal{O}(\lambda_n^{-1/(2r)}), \quad (\text{EC.4})$$

where the first inequality uses (EC.3), and the last step follows from Lemma EC.10. By Lemma EC.9 and (EC.2),

$$\hat{\mathbf{T}}_{\lambda_n}^{(a)} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{a_i = a\} \tilde{X}_i \otimes \tilde{X}_i + \lambda_n \mathbf{1} \succeq \frac{1}{2} \mathbf{T}_{\lambda_n}^{(a)} \succeq \frac{1}{2m} \mathbf{T}_{\lambda_n}^{*(a)}$$

with probability at least $1 - \delta$ whenever $n \geq c \lambda_n^{-1/(2r)} \log(1/\delta)$. This completes the proof.

EC.2. Proof of Lemma 3

The proof decomposes the prediction error into a bias term and a variance term, then bounds each using the empirical operator dominance from Lemma 2.

Fix any $a \in [m]$. Let

$$\tilde{X} = \mathbf{L}_{K^{1/2}} X, \forall X \in \mathcal{L}^2[0, 1].$$

Also, let $\gamma^{(a)} = (\mathbf{L}_{K^{1/2}})^{-1} \beta^{(a)}$, which by Assumption 1 should be a function in $\mathcal{L}^2[0, 1]$ space. Similarly, let $\hat{\gamma}^{(a)} = (\mathbf{L}_{K^{1/2}})^{-1} \hat{\beta}^{(a)} \in \mathcal{L}^2[0, 1]$.

By (5), we can write the explicit form of $\hat{\gamma}^{(a)}$:

$$\hat{\gamma}^{(a)} = \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \left(\hat{\mathbf{T}}_n^{(a)} \gamma^{(a)} + g_n^{(a)} \right), \quad (\text{EC.5})$$

where

$$\hat{\mathbf{T}}_n^{(a)} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{a_i = a\} \tilde{X}_i \otimes \tilde{X}_i \quad \text{and} \quad g_n^{(a)} = \frac{1}{n} \sum \varepsilon_i \mathbb{1}\{a_i = a\} \tilde{X}_i.$$

We now bound the right hand side of (3) by the sum of a bias term and a variance term:

$$\begin{aligned} \left| \langle X, \hat{\beta}^{(a)} \rangle - \langle X, \beta^{(a)} \rangle \right| &= \left| \langle \tilde{X}, \hat{\gamma}^{(a)} - \gamma^{(a)} \rangle \right| \\ &= \left| \langle \tilde{X}, \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \left(\hat{\mathbf{T}}_n^{(a)} \gamma^{(a)} + g_n^{(a)} \right) - \gamma^{(a)} \rangle \right| \\ &\leq \underbrace{\left| \langle \tilde{X}, \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \hat{\mathbf{T}}_n^{(a)} \gamma^{(a)} - \gamma^{(a)} \rangle \right|}_{:= b_n} + \underbrace{\left| \langle \tilde{X}, \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} g_n^{(a)} \rangle \right|}_{:= v_n}. \end{aligned} \quad (\text{EC.6})$$

Analysis of the bias term. We first analyze the term b_n . By Assumption 2, we have $\tilde{X}$ is $\tilde{\Sigma}$ -subgaussian, and thus b_n is also sub-gaussian with variance proxy

$$\left\langle \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \hat{\mathbf{T}}_n^{(a)} \gamma^{(a)} - \gamma^{(a)}, \tilde{\Sigma} \left(\left(\hat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \hat{\mathbf{T}}_n^{(a)} \gamma^{(a)} - \gamma^{(a)} \right) \right\rangle$$

$$\begin{aligned}
&= \left\langle \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \left(\widehat{\mathbf{T}}_n^{(a)} - \widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right) \gamma^{(a)}, \widetilde{\Sigma} \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \left(\widehat{\mathbf{T}}_n^{(a)} - \widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right) \gamma^{(a)} \right\rangle \\
&= \lambda_n^2 \left\langle \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \gamma^{(a)}, \widetilde{\Sigma} \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \gamma^{(a)} \right\rangle.
\end{aligned} \tag{EC.7}$$

Recall the regularized optimal-region operator $\mathbf{T}_{\lambda_n}^{*(a)} = \mathbf{L}_{M^{(a)}} + \lambda_n \mathbf{1}$. Lemma condition with Assumption 4 gives

$$\widetilde{\Sigma} \preceq c_w \mathbf{T}_{\lambda_n}^{*(a)} \preceq 2mc_w \widehat{\mathbf{T}}_{\lambda_n}^{(a)},$$

and thus the proxy term in (EC.7)

$$\begin{aligned}
\lambda_n^2 \left\langle \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \gamma^{(a)}, \widetilde{\Sigma} \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \gamma^{(a)} \right\rangle &\leq 2mc_w \lambda_n^2 \left\langle \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \gamma^{(a)}, \widehat{\mathbf{T}}_{\lambda_n}^{(a)} \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \gamma^{(a)} \right\rangle \\
&= 2mc_w \lambda_n^2 \left\langle \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \gamma^{(a)}, \gamma^{(a)} \right\rangle \\
&\leq 2mc_w \lambda_n^2 \left\| \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \right\|_{\text{op}} \|\gamma^{(a)}\|_2^2 \leq 2mc_w \lambda_n
\end{aligned}$$

where the last step uses $\|(\widehat{\mathbf{T}}_{\lambda_n}^{(a)})^{-1}\|_{\text{op}} \leq \lambda_n^{-1}$ and $\|\gamma^{(a)}\|_2 \leq 1$ from Assumption 1. Since b_n is subgaussian with variance proxy bounded by $2mc_w \lambda_n$, by Lemma EC.5 we have

$$|b_n| \leq \sqrt{4mc_w \lambda_n \log(1/\delta)}$$

with probability greater than $1 - \delta$ for any $\delta \in (0, 1)$.

Analysis of the variance term. Now we upper bound the variance term in (EC.6). We have

$$\begin{aligned}
v_n &= \langle \widetilde{X}, \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} g_n^{(a)} \rangle \\
&= \left\langle \widetilde{X}, \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbb{1}\{a_i = a\} \widetilde{X}_i \right\rangle \\
&= \frac{1}{n} \sum_{i=1}^n \varepsilon_i \left\langle \widetilde{X}, \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \widetilde{X}_i \mathbb{1}\{a_i = a\} \right\rangle.
\end{aligned} \tag{EC.8}$$

Since ε_i are i.i.d. σ_ε^2 -subgaussian noise, by Lemma EC.6, we have with probability greater than $1 - \delta$

$$\begin{aligned}
|v_n| &\leq \frac{\sigma_\varepsilon \sqrt{2 \log(1/\delta)}}{n} \sqrt{\sum_{i=1}^n \left\langle \widetilde{X}, \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \widetilde{X}_i \mathbb{1}\{a_i = a\} \right\rangle^2} \\
&= \sigma_\varepsilon \sqrt{\frac{2 \log(1/\delta)}{n}} \sqrt{\frac{\sum_{i=1}^n \left\langle \widetilde{X}, \left(\widehat{\mathbf{T}}_{\lambda_n}^{(a)} \right)^{-1} \widetilde{X}_i \mathbb{1}\{a_i = a\} \right\rangle^2}{n}}
\end{aligned}$$

$$\begin{aligned}
&= \sigma_\varepsilon \sqrt{\frac{2\log(1/\delta)}{n}} \sqrt{\left\langle \tilde{X}, \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)}\right)^{-1} \hat{\mathbf{T}}_n^{(a)} \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)}\right)^{-1} \tilde{X} \right\rangle} \\
&\leq \sigma_\varepsilon \sqrt{\frac{2\log(1/\delta)}{n}} \sqrt{\left\langle \tilde{X}, \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)}\right)^{-1} \tilde{X} \right\rangle},
\end{aligned}$$

where the last inequality follows from $\hat{\mathbf{T}}_n^{(a)} \preceq \hat{\mathbf{T}}_{\lambda_n}^{(a)}$.

Combining bias and variance. Substituting the bias and variance bounds into the decomposition (EC.6), we obtain that with probability greater than $1 - 2\delta$,

$$\left| \langle X, \hat{\beta}^{(a)} \rangle - \langle X, \beta^{(a)} \rangle \right| \leq |b_n| + |v_n| \leq 2\sqrt{2mc_w \lambda_n \log(1/\delta)} + \sigma_\varepsilon \sqrt{\frac{2\log(1/\delta)}{n} \left\langle \tilde{X}, \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)}\right)^{-1} \tilde{X} \right\rangle}$$

for all $a \in [m]$. Recalling that $\tilde{X} = \mathbf{L}_{K^{1/2}} X$ and rescaling δ by a constant factor, we conclude that (3) holds with

$$\Delta^{(a)}(X) \asymp \sqrt{\lambda_n \log(1/\delta)} + \sqrt{\frac{\log(1/\delta)}{n} \langle \mathbf{L}_{K^{1/2}} X, \left(\hat{\mathbf{T}}_{\lambda_n}^{(a)}\right)^{-1} \mathbf{L}_{K^{1/2}} X \rangle}$$

with probability greater than $1 - \delta$ for all X and $a \in [m]$. This completes the proof.

EC.3. Proof of Theorem 1

Define the *good event*

$$\mathcal{E} = \bigcap_{\tau=1}^{\tau_0} \bigcap_{a \in [m]} \left\{ \left| \langle X, \hat{\beta}_\tau^{(a)} \rangle - \langle X, \beta^{(a)} \rangle \right| \leq \Delta_\tau^{(a)}(X), \forall X \in \mathcal{X} \right\},$$

on which the oracle output is valid in every epoch simultaneously. By the oracle guarantee (3) with $\delta = m^{-1}T^{-3}$ and a union bound over the $\tau_0 = \mathcal{O}(\log T)$ epochs, $\mathbb{P}(\mathcal{E}) \geq 1 - m\tau_0\delta \geq 1 - T^{-2}$.

We first note that the optimal action is never eliminated for any $t \in [T]$ on $\mathcal{E}$. Fix any epoch τ and any time step t in epoch τ . On $\mathcal{E}$, for any $a' \in [m]$,

$$\langle X_t, \hat{\beta}_\tau^{(a^*)} \rangle + \Delta_\tau^{(a^*)}(X_t) \geq \langle X_t, \beta^{(a^*)} \rangle \geq \langle X_t, \beta^{(a')} \rangle \geq \langle X_t, \hat{\beta}_\tau^{(a')} \rangle - \Delta_\tau^{(a')}(X_t),$$

where the first and third inequalities use the oracle validity on $\mathcal{E}$ and the second uses optimality of a^* . Hence a^* is never eliminated by Algorithm 1, i.e., $a^* \in \mathcal{A}_t$ for all $t \in [T]$.

Now we are ready to show the per-step regret bound on $\mathcal{E}$. Fix epoch $\tau \geq 2$ and time step t in epoch τ . If $a_t = a^*(X_t)$, the regret is zero. If $a_t \neq a^*(X_t)$, then $a_t \in \mathcal{A}_t$ means it is not eliminated, so

$$\langle X_t, \hat{\beta}_\tau^{(a_t)} \rangle + \Delta_\tau^{(a_t)}(X_t) \geq \max_{a' \in [m]} \langle X_t, \hat{\beta}_\tau^{(a')} \rangle - \Delta_\tau^{(a')}(X_t) \geq \langle X_t, \hat{\beta}_\tau^{(a^*)} \rangle - \Delta_\tau^{(a^*)}(X_t).$$

On $\mathcal{E}$, the left-hand side is at most $\langle X_t, \beta^{(a_t)} \rangle + 2\Delta_\tau^{(a_t)}(X_t)$ and the right-hand side is at least $\langle X_t, \beta^{(a^*)} \rangle - 2\Delta_\tau^{(a^*)}(X_t)$. Therefore

$$0 < \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \leq 2\Delta_\tau^{(a_t)}(X_t) + 2\Delta_\tau^{(a^*(X_t))}(X_t). \quad (\text{EC.9})$$

With the per-period regret bound, the total regret decomposes as

$$\mathbb{E}[\text{Reg}_{\text{Alg. 1}}(T)] = \underbrace{\mathbb{E}\left[\sum_{t=1}^T \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}}\right]}_{(\text{I})} + \underbrace{\mathbb{E}\left[\sum_{t=1}^T \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}^c}\right]}_{(\text{II})}. \quad (\text{EC.10})$$

For term (II) in (EC.10), since for any $a \in [m]$, $|\langle X_t, \beta^{(a)} \rangle| = |\langle \tilde{X}_t, \gamma^{(a)} \rangle| \leq \|\tilde{X}_t\|$ by $\|\gamma^{(a)}\|_2 \leq 1$ (Assumption 1) and $\mathbb{E}[\|\tilde{X}_t\|^2] = \text{Tr}(\tilde{\Sigma}) < \infty$ (since $r > 1/2$), Cauchy–Schwarz gives

$$\mathbb{E}\left[\sum_{t=1}^T \left| \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \right| \mathbb{1}_{\mathcal{E}^c}\right] \leq 2T \sqrt{\text{Tr}(\tilde{\Sigma})} \sqrt{\mathbb{P}(\mathcal{E}^c)} = \mathcal{O}(1),$$

given that $\mathbb{P}(\mathcal{E}^c) \leq T^{-2}$.

For term (I) in (EC.10), we similarly have

$$\begin{aligned} (\text{I}) &= \sum_{\tau=2}^{\tau_0} \sum_{t \in \mathcal{T}_\tau} \mathbb{E}\left[\langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}}\right] + \sum_{t \in \mathcal{T}_1} \mathbb{E}\left[\langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}}\right] \\ &= \sum_{\tau=2}^{\tau_0} \sum_{t \in \mathcal{T}_\tau} \mathbb{E}\left[\langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}}\right] + \mathcal{O}(1). \end{aligned} \quad (\text{EC.11})$$

On $\mathcal{E}$, the per-step regret is zero when $a_t = a^*(X_t)$. When $a_t \neq a^*(X_t)$, since $a_t \in \mathcal{A}_t$, (EC.9) gives $0 < \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \leq 2\Delta_\tau^{(a_t)}(X_t) + 2\Delta_\tau^{(a^*(X_t))}(X_t)$, so the gap indicator $\mathbb{1}\{0 < \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle < 2\Delta_\tau^{(a_t)}(X_t) + 2\Delta_\tau^{(a^*(X_t))}(X_t)\}$ equals 1. Therefore, the summand in (EC.11) has

$$\begin{aligned} &\langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}} \\ &\leq 2(\Delta_\tau^{(a_t)}(X_t) + \Delta_\tau^{(a^*(X_t))}(X_t)) \mathbb{1}\{0 < \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle < 2\Delta_\tau^{(a_t)}(X_t) + 2\Delta_\tau^{(a^*(X_t))}(X_t)\}. \end{aligned} \quad (\text{EC.12})$$

The oracle output $(\hat{\beta}_\tau^{(a)}, \Delta_\tau^{(a)})$ depends only on epoch $\tau - 1$ data, so $X_t \sim \mathcal{F}$ is independent of $(\hat{\beta}_\tau^{(a)}, \Delta_\tau^{(a)})$ for each t in epoch τ . Conditioning on X_t and the oracle output, the uniform action selection $a_t \sim \text{Uniform}(\mathcal{A}_t)$ gives

$$\mathbb{E}_{a_t} \left[\langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}} \mid X_t, \hat{\beta}_\tau, \Delta_\tau \right] = \frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t} \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a)} \rangle \mathbb{1}_{\mathcal{E}}. \quad (\text{EC.13})$$

Substituting (EC.12) into each summand of (EC.13) and relaxing $\frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t}$ to $\sum_{a \in [m]}$, we obtain

$$\begin{aligned} & \mathbb{E}_{a_t} \left[\langle X_t, \beta^{(a^*(X_t))} - \beta^{(a_t)} \rangle \mathbb{1}_{\mathcal{E}} \mid X_t, \widehat{\beta}_\tau, \Delta_\tau \right] \\ & \leq 2 \sum_{a \in [m]} (\Delta_\tau^{(a)}(X_t) + \Delta_\tau^{(a^*(X_t))}(X_t)) \cdot \mathbb{1}_{\{0 < \left| \langle X_t, \beta^{(a^*(X_t))} - \beta^{(a)} \rangle \right| < 2\Delta_\tau^{(a)}(X_t) + 2\Delta_\tau^{(a^*(X_t))}(X_t)\}}. \end{aligned} \quad (\text{EC.14})$$

Taking expectation over both X_t and the oracle randomness, summing over the $|\mathcal{T}_\tau| = 2^\tau$ time steps in epoch τ and over $\tau = 2, \dots, \tau_0$, substituting into (EC.11) gives

$$\begin{aligned} (\text{I}) & \leq \sum_{\tau=2}^{\tau_0} 2^\tau \cdot 2 \sum_{a \in [m]} \mathbb{E} \left[(\Delta_\tau^{(a)}(X) + \Delta_\tau^{(a^*(X))}(X)) \right. \\ & \quad \left. \cdot \mathbb{1}_{\{0 < \left| \langle X, \beta^{(a^*(X))} - \beta^{(a)} \rangle \right| < 2\Delta_\tau^{(a)}(X) + 2\Delta_\tau^{(a^*(X))}(X)\}} \right] + \mathcal{O}(1). \end{aligned} \quad (\text{EC.15})$$

Combining (EC.15) for term (I) and the $\mathcal{O}(1)$ bound for term (II) in (EC.10) completes the proof.

EC.4. Proof of Theorem 2

We first show that the oracle $\mathcal{O}$ is valid across all epochs by induction on $\tau = 1, \dots, \tau_0$. In epoch $\tau = 1$, the algorithm has no prior samples, so (4) is satisfied by Lemma 1. Now suppose that for all $a \in [m]$, the oracle output $(\widehat{\beta}_{\tau-1}^{(a)}, \Delta_{\tau-1}^{(a)})$ in epoch $\tau - 1$ satisfies (3). Since $\lambda_T \leq \lambda_{2^{\tau-1}}$ for all τ , Assumption 4 holds with $\mu = \lambda_{2^{\tau-1}}$ as required by Lemma 2. To apply Lemma 2, we also need $(c \log(T/\delta)/n)^{2r} \leq \lambda_n$. With our choice $\lambda_n \asymp (\log(mT)/n)^{2r/(2r+1)}$, by choosing the implicit constant sufficiently large,

$$\left(\frac{c \log(T/\delta)}{n} \right)^{2r} \leq \lambda_n.$$

By Lemma 2 applied to the $n = 2^{\tau-1}$ samples $\{(X_i, a_i, y_i)\}$ collected in epoch $\tau - 1$, the empirical dominance

$$\frac{1}{2m} \mathbf{T}_{\lambda_{2^{\tau-1}}}^{*(a)} \preceq \widehat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)}$$

holds with probability at least $1 - \delta/T$ for all $a \in [m]$. Given this dominance, Lemma 3 then guarantees that for all $a \in [m]$, the oracle output $(\widehat{\beta}_\tau^{(a)}, \Delta_\tau^{(a)})$ in epoch τ satisfies (3) with probability at least $1 - \delta/T$. By induction and a union bound over the $\tau_0 = \mathcal{O}(\log T)$ epochs (accounting for both the dominance and oracle failure in each epoch), the oracle is valid in every epoch simultaneously with probability at least $1 - 2\tau_0\delta/T \geq 1 - \delta$.

By Theorem 1, dropping the indicator and noting $\sum_{a \in [m]} \Delta_\tau^{(a^*(X))}(X) = m \Delta_\tau^{(a^*(X))}(X) \leq m \sum_{a \in [m]} \Delta_\tau^{(a)}(X)$, we have

$$\mathbb{E} [\text{Reg}_{\text{Alg. 1}}(T)] \leq \mathcal{O} \left(\sum_{\tau=2}^{\tau_0} 2^\tau \sum_{a \in [m]} \mathbb{E} [\Delta_\tau^{(a)}(X)] \right) + \mathcal{O}(1).$$

Fix any epoch $\tau \in \{2, \dots, \tau_0\}$. It remains to bound $\mathbb{E}[\Delta_\tau^{(a)}(X)]$ for each $a \in [m]$, where $X \sim \mathcal{F}$ is independent of the epoch $\tau - 1$ data. By Lemma 3 with $n = 2^{\tau-1}$ and $\delta = m^{-1}T^{-3}$, the confidence radius satisfies

$$\Delta_\tau^{(a)}(X) \preceq \sqrt{\lambda_{2^{\tau-1}} \log(mT)} + \sqrt{\frac{\log(mT)}{2^{\tau-1}} \langle \mathbf{L}_{K^{1/2}} X, \left(\widehat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)} \right)^{-1} \mathbf{L}_{K^{1/2}} X \rangle}. \quad (\text{EC.16})$$

Let $\mathcal{D}_\tau$ denote the event that $\widehat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)} \succeq \frac{1}{2m} \mathbf{T}_{\lambda_{2^{\tau-1}}}^{*(a)}$ holds for all $a \in [m]$; by Lemma 2, $\mathbb{P}(\mathcal{D}_\tau^c) \leq T^{-2}$. We split each summand:

$$\mathbb{E} [\Delta_\tau^{(a)}(X)] = \mathbb{E} [\Delta_\tau^{(a)}(X) \mathbb{1}_{\mathcal{D}_\tau}] + \mathbb{E} [\Delta_\tau^{(a)}(X) \mathbb{1}_{\mathcal{D}_\tau^c}].$$

For the on-event term, by Assumption 4, on $\mathcal{D}_\tau$ we have

$$\widehat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)} \succeq \frac{1}{2m} \mathbf{T}_{\lambda_{2^{\tau-1}}}^{*(a)} \succeq \frac{1}{2c_w m} (\tilde{\mathbf{\Sigma}} + \lambda_{2^{\tau-1}} \mathbf{1})$$

for all $a \in [m]$. Substituting into (EC.16),

$$\begin{aligned} \mathbb{E} [\Delta_\tau^{(a)}(X) \mathbb{1}_{\mathcal{D}_\tau}] &\leq \mathcal{O} \left(\sqrt{\lambda_{2^{\tau-1}} \log(mT)} + \sqrt{\frac{\log(mT)}{2^{\tau-1}}} \mathbb{E} \left[\sqrt{\langle \tilde{X}, (\tilde{\mathbf{\Sigma}} + \lambda_{2^{\tau-1}} \mathbf{1})^{-1} \tilde{X} \rangle} \right] \right) \\ &\leq \mathcal{O} \left(\sqrt{\lambda_{2^{\tau-1}} \log(mT)} + \sqrt{\frac{\log(mT)}{2^{\tau-1}}} \sqrt{\text{Tr}(\tilde{\mathbf{\Sigma}} (\tilde{\mathbf{\Sigma}} + \lambda_{2^{\tau-1}} \mathbf{1})^{-1})} \right) \\ &\leq \mathcal{O} \left(\sqrt{\lambda_{2^{\tau-1}} \log(mT)} + \sqrt{\frac{\log(mT)}{2^{\tau-1}}} (\lambda_{2^{\tau-1}})^{-\frac{1}{4r}} \right), \end{aligned}$$

where the second line uses Jensen's inequality and the trace identity, and the third line applies Lemma EC.10.

For the off-event term, since $\widehat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)} \succeq \lambda_{2^{\tau-1}} \mathbf{1}$ by definition, $(\widehat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)})^{-1} \preceq (\lambda_{2^{\tau-1}})^{-1} \mathbf{1}$. By independence of X and the oracle output, (EC.16) gives

$$\mathbb{E}[(\Delta_\tau^{(a)}(X))^2] \lesssim \lambda_{2^{\tau-1}} \log(mT) + \frac{\log(mT)}{2^{\tau-1} \lambda_{2^{\tau-1}}} \text{Tr}(\tilde{\mathbf{\Sigma}}) = \mathcal{O}(1),$$

since $\text{Tr}(\tilde{\mathbf{\Sigma}}) = \mathcal{O}(1)$ by $r > 1/2$ and $\lambda_{2^{\tau-1}} \geq \Omega((2^{\tau-1})^{-1})$. Cauchy-Schwarz gives

$$\mathbb{E} [\Delta_\tau^{(a)}(X) \mathbb{1}_{\mathcal{D}_\tau^c}] \leq \sqrt{\mathbb{E}[(\Delta_\tau^{(a)}(X))^2]} \cdot \sqrt{\mathbb{P}(\mathcal{D}_\tau^c)} = \mathcal{O}(T^{-1}).$$

Combining the two contributions and substituting into Theorem 1, noting $\sum_{\tau=2}^{\tau_0} 2^\tau \cdot \mathcal{O}(T^{-1}) = \mathcal{O}(1)$, we obtain

$$\mathbb{E}[\text{Reg}_{\text{Alg. 1}}(T)] \leq \sum_{\tau=2}^{\tau_0} 2^\tau \mathcal{O} \left(\sqrt{\lambda_{2^{\tau-1}} \log(mT)} + \sqrt{\frac{\log(mT)}{2^{\tau-1}}} (\lambda_{2^{\tau-1}})^{-\frac{1}{4r}} \right).$$

Now, substituting the choice $\lambda_n \asymp (\log(mT)/n)^{2r/(2r+1)}$, a direct computation shows that both

$$\sqrt{\lambda_{2^{\tau-1}} \log(mT)} = \tilde{\mathcal{O}}\left(2^{-\frac{r(\tau-1)}{2r+1}}\right) \quad \text{and} \quad \sqrt{\frac{\log(mT)}{2^{\tau-1}}} \cdot (\lambda_{2^{\tau-1}})^{-\frac{1}{4r}} = \tilde{\mathcal{O}}\left(2^{-\frac{r(\tau-1)}{2r+1}}\right).$$

Substituting into the above display, each epoch τ contributes $2^\tau \cdot \tilde{\mathcal{O}}(2^{-\frac{r(\tau-1)}{2r+1}}) = \tilde{\mathcal{O}}(2^{\frac{\tau(r+1)}{2r+1}})$ to the regret. Summing this geometric series over $\tau = 2, \dots, \tau_0$,

$$\mathbb{E}[\text{Reg}_{\text{Alg. 1}}(T)] \leq \sum_{\tau=2}^{\tau_0} \tilde{\mathcal{O}}\left(2^{\frac{\tau(r+1)}{2r+1}}\right) + \mathcal{O}(1) = \tilde{\mathcal{O}}\left(2^{\frac{\tau_0(r+1)}{2r+1}}\right).$$

Since $2^{\tau_0} = \mathcal{O}(T)$ by the geometric design of Algorithm 1, this yields the final result.

EC.5. Proof of Theorem 4

Fix any epoch $\tau \in \{2, \dots, \tau_0\}$. Recall that from Lemma 3 for any $a \in [m]$

$$\Delta_\tau^{(a)}(X) \asymp \sqrt{\lambda_{2^{\tau-1}} \log(mT)} + \sqrt{\frac{\log(mT)}{2^{\tau-1}} \langle \mathbf{L}_{K^{1/2}} X, \left(\hat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)}\right)^{-1} \mathbf{L}_{K^{1/2}} X \rangle}. \quad (\text{EC.17})$$

The following lemma provides a high probability bound for the variance term in $\Delta_\tau^{(a)}(X)$.

LEMMA EC.1. *For any $a \in [m]$, with probability higher than $1 - 2m^{-1}T^{-2}$,*

$$\sqrt{\langle \mathbf{L}_{K^{1/2}} X, \left(\hat{\mathbf{T}}_{\lambda_{2^{\tau-1}}}^{(a)}\right)^{-1} \mathbf{L}_{K^{1/2}} X \rangle} \leq 2(\lambda_{2^{\tau-1}})^{-\frac{1}{4r}} + \log(mT).$$

Let $\mathcal{B}_\tau$ denote the event that the bound in Lemma EC.1 holds simultaneously for all $a \in [m]$.

Substituting Lemma EC.1 and the choice of $\lambda_{2^{\tau-1}} \asymp (\log(mT)/2^{\tau-1})^{2r/(2r+1)}$ into (EC.17), on $\mathcal{B}_\tau$ we obtain the deterministic bound $\Delta_\tau^{(a)}(X) \lesssim \bar{\Delta}_\tau$ for all $a \in [m]$, where

$$\bar{\Delta}_\tau := \sqrt{\lambda_{2^{\tau-1}} \log(mT)} + \sqrt{\frac{\log(mT)}{2^{\tau-1}}} \left((\lambda_{2^{\tau-1}})^{-\frac{1}{4r}} + \log(mT) \right) = \tilde{\mathcal{O}}\left(2^{-\frac{r(\tau-1)}{2r+1}}\right).$$

We split into on-event and off-event contributions. On $\mathcal{B}_\tau$, we have $\Delta_\tau^{(a)}(X) \leq \bar{\Delta}_\tau$ for all $a \in [m]$.

Off $\mathcal{B}_\tau$, Cauchy–Schwarz gives $\mathbb{E}[\Delta_\tau^{(a)}(X) \mathbb{1}_{\mathcal{B}_\tau^c}] \leq \sqrt{\mathbb{E}[(\Delta_\tau^{(a)}(X))^2]} \cdot \sqrt{\mathbb{P}(\mathcal{B}_\tau^c)}$, which is $\mathcal{O}(T^{-1})$ since $\mathbb{E}[(\Delta_\tau^{(a)}(X))^2]$ is bounded and $\mathbb{P}(\mathcal{B}_\tau^c) \leq 2T^{-2}$. Therefore,

$$\begin{aligned} & \sum_{a \in [m]} \mathbb{E} \left[(\Delta_\tau^{(a)}(X) + \Delta_\tau^{(a^*(X))}(X)) \mathbb{1} \left\{ 0 < \left| \langle X, \beta^{(a^*(X))} - \beta^{(a)} \rangle \right| < 2\Delta_\tau^{(a)}(X) + 2\Delta_\tau^{(a^*(X))}(X) \right\} \right] \\ & \leq \sum_{a \in [m]} \mathbb{E} \left[2\bar{\Delta}_\tau \mathbb{1} \left\{ 0 < \left| \langle X, \beta^{(a^*(X))} - \beta^{(a)} \rangle \right| < 4\bar{\Delta}_\tau \right\} \mathbb{1}_{\mathcal{B}_\tau} \right] + \mathcal{O}(T^{-1}) \\ & \leq 2mc_m(4\bar{\Delta}_\tau)^{\alpha+1} + \mathcal{O}(T^{-1}), \end{aligned}$$

where the last inequality applies the margin condition in Assumption 5. Since $2 \cdot 4^{\alpha+1}$ is a constant, we absorb it into $\mathcal{O}(\cdot)$.

To get our final result, note that by Theorem 1 we have

$$\begin{aligned} \mathbb{E} [\text{Reg}_{\text{Alg. 1}}(T)] &\leq \mathcal{O} \left(\sum_{\tau=2}^{\tau_0} 2^\tau \sum_{a \in [m]} \mathbb{E} \left[(\Delta_\tau^{(a)}(X) + \Delta_\tau^{(a^*(X))}(X)) \right. \right. \\ &\quad \cdot \mathbb{1} \left\{ 0 < \left| \langle X, \beta^{(a^*(X))} - \beta^{(a)} \rangle \right| < 2\Delta_\tau^{(a)}(X) + 2\Delta_\tau^{(a^*(X))}(X) \right\} \left. \right] \Bigg) + \mathcal{O}(1) \\ &\leq \mathcal{O} \left(\sum_{\tau=2}^{\tau_0} 2^\tau \bar{\Delta}_\tau^{\alpha+1} \right) + \mathcal{O}(1). \end{aligned}$$

Substituting $\bar{\Delta}_\tau = \tilde{\mathcal{O}}(2^{-r(\tau-1)/(2r+1)})$ gives

$$\sum_{\tau=2}^{\tau_0} 2^\tau \bar{\Delta}_\tau^{\alpha+1} = \sum_{\tau=2}^{\tau_0} 2^\tau \tilde{\mathcal{O}} \left(2^{-\frac{r(\alpha+1)(\tau-1)}{2r+1}} \right) = \sum_{\tau=2}^{\tau_0} \tilde{\mathcal{O}} \left(2 \cdot 2^{(\tau-1)(1-\frac{r(\alpha+1)}{2r+1})} \right) = \sum_{\tau=2}^{\tau_0} \tilde{\mathcal{O}} \left(2^{\frac{(\tau-1)(r+1-r\alpha)}{2r+1}} \right).$$

Since $\alpha < 1 + 1/r$, we have $r + 1 - r\alpha > 0$, so each term grows geometrically in τ and the sum is dominated by the last term $\tau = \tau_0$:

$$\sum_{\tau=2}^{\tau_0} \tilde{\mathcal{O}} \left(2^{\frac{(\tau-1)(r+1-r\alpha)}{2r+1}} \right) = \tilde{\mathcal{O}} \left(2^{\frac{(\tau_0-1)(r+1-r\alpha)}{2r+1}} \right) = \tilde{\mathcal{O}} \left(2^{\frac{\tau_0(r+1-r\alpha)}{2r+1}} \right).$$

Since $2^{\tau_0} = \mathcal{O}(T)$ by our geometric design, we conclude

$$\mathbb{E} [\text{Reg}_{\text{Alg. 1}}(T)] \leq \tilde{\mathcal{O}} \left(T^{\frac{r+1-r\alpha}{2r+1}} \right).$$

EC.6. Proof of Theorem 5

We begin by establishing empirical operator dominance for all models $s \leq s^*$, which guarantees that the optimal action is never eliminated. Under this good event $\mathcal{E}_t$, the per-step regret is controlled by the confidence width of the true model s^* . We then bound the expected confidence width via the effective dimension to obtain the final rate.

LEMMA EC.2. *Under the assumptions of Theorem 5 we have*

$$\hat{\mathbf{T}}_{s,t}^{(a)} \succeq \frac{1}{4c_o} \left(\tilde{\Sigma}_s + \lambda_{s,t} \mathbf{1} \right), \quad \text{for all } s \leq s^*, a \in [m], \text{ and } t \geq 2t_0, \quad (\text{EC.18})$$

with probability at least $1 - 4T^{-2}$, where $c_o > 0$ is the constant from Assumption 8.

With this dominance, the oracle output is valid for all models $s \leq s^*$, so that the optimal action survives elimination.

LEMMA EC.3. *Instate the assumptions in Theorem 5. If the empirical dominance (EC.18) holds, then*

$$\left| \langle X_t, \widehat{\beta}_{s,t}^{(a)} \rangle - \langle X_t, \beta^{(a)} \rangle \right| \leq \Delta_{s,t}^{(a)}, \quad \text{for all } s \leq s^*, a \in [m], \text{ and } t \geq 2t_0,$$

with probability at least $1 - 3T^{-2}$.

For any $t \geq 2t_0$, let $\mathcal{E}_t$ denote the event that (EC.18) and the oracle validity from Lemma EC.3 hold simultaneously. On $\mathcal{E}_t$, for every $s \leq s^*$ and any $a' \in [m]$,

$$\langle X_t, \widehat{\beta}_{s,t}^{(a^*)} \rangle + \Delta_{s,t}^{(a^*)} \geq \langle X_t, \beta^{(a^*)} \rangle \geq \langle X_t, \beta^{(a')} \rangle \geq \langle X_t, \widehat{\beta}_{s,t}^{(a')} \rangle - \Delta_{s,t}^{(a')},$$

so $a^*(X_t) \in \mathcal{A}_t^{s^*}$. Since $\bigcap_{s=1}^{s^*} \mathcal{A}_t^s \neq \emptyset$, the algorithm selects $s_t \geq s^*$, and hence $a_t \in \bigcap_{s=1}^{s_t} \mathcal{A}_t^s \subseteq \mathcal{A}_t^{s^*}$.

Now, suppose that $\mathcal{E}_t$ holds for every $t \geq 2t_0$, we can decompose the cumulative regret from time period $2t_0$ to time period T as

$$\begin{aligned} \sum_{t=2t_0}^T \langle X_t, \beta^{(a^*(X_t))} \rangle - \langle X_t, \beta^{(a_t)} \rangle &\leq \sum_{t=2t_0}^T \max_{a \in \mathcal{A}_t^{s^*}} \langle X_t, \beta^{(a^*(X_t))} \rangle - \langle X_t, \beta^{(a)} \rangle \\ &\leq \sum_{t=2t_0}^T \langle X_t, \widehat{\beta}_t^{(a^*(X_t))} \rangle + \Delta_{s^*,t}^{(a^*(X_t))} - \min_{a \in \mathcal{A}_t^{s^*}} \left(\langle X_t, \widehat{\beta}_t^{(a)} \rangle - \Delta_{s^*,t}^{(a)} \right) \\ &\leq \sum_{t=2t_0}^T \left(\Delta_{s^*,t}^{(a^*(X_t))} + \max_{a \in \mathcal{A}_{s^*,t}^{s^*}} \Delta_{s^*,t}^{(a)} \right) + \left(\langle X_t, \widehat{\beta}_t^{(a^*(X_t))} \rangle - \min_{a \in \mathcal{A}_t^{s^*}} \langle X_t, \widehat{\beta}_t^{(a)} \rangle \right) \\ &\leq \sum_{t=2t_0}^T 2 \left(\Delta_{s^*,t}^{(a^*(X_t))} + \max_{a \in \mathcal{A}_{s^*,t}^{s^*}} \Delta_{s^*,t}^{(a)} \right), \end{aligned}$$

where the last inequality is given by

$$\min_{a \in \mathcal{A}_t^{s^*}} \langle X_t, \widehat{\beta}_t^{(a)} \rangle + \max_{a \in \mathcal{A}_t^{s^*}} \Delta_{s^*,t}^{(a)} \geq \min_{a \in \mathcal{A}_t^{s^*}} \langle X_t, \widehat{\beta}_t^{(a)} \rangle + \Delta_{s^*,t}^{(a)} \geq \max_{a \in \mathcal{A}_t^{s^*}} \langle X_t, \widehat{\beta}_t^{(a)} \rangle - \Delta_{s^*,t}^{(a)} \geq \langle X_t, \widehat{\beta}_t^{(a^*(X_t))} \rangle - \Delta_{s^*,t}^{(a^*(X_t))},$$

due to our elimination rule in the algorithm and that $a^*(X_t) \in \mathcal{A}_t^{s^*}$. We now proceed with

$$\sum_{t=2t_0}^T 2 \left(\Delta_{s^*,t}^{(a^*(X_t))} + \max_{a \in \mathcal{A}_{s^*,t}^{s^*}} \Delta_{s^*,t}^{(a)} \right) \leq \sum_{t=2t_0}^T 4 \max_{a \in [m]} \Delta_{s^*,t}^{(a)}$$

and take expectation and account for the event $\mathcal{E}_t$ not holding. Since the per-step regret satisfies $|\langle X_t, \beta^{(a^*(X_t))} \rangle - \langle X_t, \beta^{(a_t)} \rangle| \leq 2\|\tilde{X}_t\|$ (by $\|\gamma^{(a)}\|_2 \leq 1$ from Assumption 1) and $\mathbb{E}[\|\tilde{X}_t\|^2] = \text{Tr}(\tilde{\Sigma}) < \infty$, Cauchy–Schwarz gives

$$\mathbb{E} [\text{Regret}_{\text{Alg. 2}}(T)] = \mathbb{E} \left[\sum_{t=1}^{2t_0-1} \left(\langle X_t, \beta^{(a^*(X_t))} \rangle - \langle X_t, \beta^{(a_t)} \rangle \right) \right] + \sum_{t=2t_0}^T \mathbb{E} \left[\langle X_t, \beta^{(a^*(X_t))} \rangle - \langle X_t, \beta^{(a_t)} \rangle \right]$$

$$\begin{aligned}
&\leq \mathcal{O}(t_0) + \sum_{t=2t_0}^T \left(\mathbb{E} \left[\left(\langle X_t, \beta^{(a^*(X_t))} \rangle - \langle X_t, \beta^{(at)} \rangle \right) \mathbb{1}_{\mathcal{E}_t} \right] + 2\sqrt{\text{Tr}(\tilde{\Sigma})} \sqrt{\mathbb{P}(\mathcal{E}_t^c)} \right) \\
&\leq \mathcal{O}(t_0) + 4 \sum_{t=2t_0}^T \mathbb{E} \left[\max_{a \in [m]} \Delta_{s^*,t}^{(a)} \cdot \mathbb{1}_{\mathcal{E}_t} \right] + 2\sqrt{\text{Tr}(\tilde{\Sigma})} \sum_{t=2t_0}^T \sqrt{\mathbb{P}(\mathcal{E}_t^c)},
\end{aligned}$$

where in the last inequality we used the per-period regret bound $4 \max_{a \in [m]} \Delta_{s^*,t}^{(a)}$ on $\mathcal{E}_t$ established above. By Lemma EC.2, (EC.18) holds with probability at least $1 - 4T^{-2}$. Given this, Lemma EC.3 gives the oracle validity with probability at least $1 - 3T^{-2}$. A union bound yields $\mathbb{P}(\mathcal{E}_t^c) \leq 7T^{-2}$ for each $t \geq 2t_0$, and

$$2\sqrt{\text{Tr}(\tilde{\Sigma})} \sum_{t=2t_0}^T \sqrt{\mathbb{P}(\mathcal{E}_t^c)} \leq 2\sqrt{\text{Tr}(\tilde{\Sigma})} \cdot T \cdot \sqrt{7} T^{-1} = \mathcal{O}(1),$$

and combining with $t_0 = \tilde{\mathcal{O}}(1)$ gives

$$\mathbb{E} [\text{Regret}_{\text{Alg. 2}}(T)] \leq 4 \sum_{t=2t_0}^T \mathbb{E} \left[\max_{a \in [m]} \Delta_{s^*,t}^{(a)} \cdot \mathbb{1}_{\mathcal{E}_t} \right] + \tilde{\mathcal{O}}(1). \quad (\text{EC.19})$$

It remains to bound $\mathbb{E} \left[\max_{a \in [m]} \Delta_{s^*,t}^{(a)} \cdot \mathbb{1}_{\mathcal{E}_t} \right]$. By Lemma 3 applied to model s^* with $n = t - 1$ and $\lambda_n = \lambda_{s^*,t-1}$, the confidence radius at time t satisfies

$$\Delta_{s^*,t}^{(a)}(X_t) \asymp \sqrt{\lambda_{s^*,t-1} \log(mT)} + \sqrt{\frac{\log(mT)}{t-1} \langle \mathbf{L}_{K_{s^*}^{1/2}} X_t, \left(\hat{\mathbf{T}}_{s^*,t-1}^{(a)} \right)^{-1} \mathbf{L}_{K_{s^*}^{1/2}} X_t \rangle}, \quad (\text{EC.20})$$

where $\hat{\mathbf{T}}_{s^*,t-1}^{(a)}$ is built from data $1, \dots, t-1$. On $\mathcal{E}_t$, the empirical dominance (EC.18) applied at index $t-1$ gives

$$\langle \mathbf{L}_{K_{s^*}^{1/2}} X_t, \left(\hat{\mathbf{T}}_{s^*,t-1}^{(a)} \right)^{-1} \mathbf{L}_{K_{s^*}^{1/2}} X_t \rangle \leq 4c_o \langle \mathbf{L}_{K_{s^*}^{1/2}} X_t, \left(\tilde{\Sigma}_{s^*} + \lambda_{s^*,t-1} \mathbf{1} \right)^{-1} \mathbf{L}_{K_{s^*}^{1/2}} X_t \rangle$$

for all $a \in [m]$, where the right-hand side no longer depends on a . Since $\hat{\mathbf{T}}_{s^*,t-1}^{(a)}$ depends only on data $1, \dots, t-1$, we apply Jensen's inequality $\mathbb{E}[\sqrt{Y}] \leq \sqrt{\mathbb{E}[Y]}$ and the trace identity to obtain

$$\begin{aligned}
\mathbb{E} \left[\max_{a \in [m]} \Delta_{s^*,t}^{(a)} \cdot \mathbb{1}_{\mathcal{E}_t} \right] &\leq \mathcal{O} \left(\sqrt{\lambda_{s^*,t-1} \log(mT)} + \sqrt{\frac{c_o \log(mT)}{t-1}} \cdot \mathbb{E} \left[\sqrt{\langle \mathbf{L}_{K_{s^*}^{1/2}} X_t, \left(\tilde{\Sigma}_{s^*} + \lambda_{s^*,t-1} \mathbf{1} \right)^{-1} \mathbf{L}_{K_{s^*}^{1/2}} X_t \rangle} \right] \right) \\
&\leq \mathcal{O} \left(\sqrt{\lambda_{s^*,t-1} \log(mT)} + \sqrt{\frac{c_o \log(mT)}{t-1}} \cdot \text{Tr} \left(\tilde{\Sigma}_{s^*} \left(\tilde{\Sigma}_{s^*} + \lambda_{s^*,t-1} \mathbf{1} \right)^{-1} \right) \right).
\end{aligned}$$

By Lemma EC.10 under Assumption 7, $\text{Tr}\left(\tilde{\Sigma}_{s^*}(\tilde{\Sigma}_{s^*} + \lambda_{s^*,t-1}\mathbf{1})^{-1}\right) = \mathcal{O}(\lambda_{s^*,t-1}^{-1/(2r_{s^*})})$. Substituting into (EC.19) and summing over t ,

$$\mathbb{E}[\text{Regret}_{\text{Alg. 2}}(T)] \leq \tilde{\mathcal{O}}\left(\sum_{t=2t_0}^T \sqrt{\lambda_{s^*,t-1}} + \sqrt{\frac{1}{t-1}} (\lambda_{s^*,t-1})^{-\frac{1}{4r_{s^*}}}\right) + \tilde{\mathcal{O}}(1),$$

Now, substituting the choice $\lambda_{s,t} \asymp (\log(mT)/t)^{2r_s/(2r_s+1)}$ for all $s \in [l]$ and t , a direct computation shows that $\sqrt{\lambda_{s^*,t-1}} \asymp t^{-r_{s^*}/(2r_{s^*}+1)}$ and $\sqrt{1/(t-1)} \cdot (\lambda_{s^*,t-1})^{-1/(4r_{s^*})} \asymp t^{-r_{s^*}/(2r_{s^*}+1)}$. The integral calculation gives

$$\mathcal{O}\left(\sum_{t=2t_0}^T t^{-\frac{r_{s^*}}{2r_{s^*}+1}}\right) = \tilde{\mathcal{O}}\left(T^{\frac{r_{s^*}+1}{2r_{s^*}+1}}\right).$$

Substituting into (EC.19), we conclude

$$\mathbb{E}[\text{Regret}_{\text{Alg. 2}}(T)] \leq \tilde{\mathcal{O}}\left(T^{\frac{r_{s^*}+1}{2r_{s^*}+1}}\right).$$

This completes the proof.

EC.6.1. Proof of Lemma EC.2

The proof relies on the following lemma, which shows that the exploration phase produces sufficiently accurate estimates for all models $s \leq s^*$.

LEMMA EC.4. *Instate the assumptions in Theorem 5. With $\lambda_{s,t_0} \asymp (\log(mT)/t_0)^{2r_s/(2r_s+1)}$ for all $s \in [l]$, after $t_0 = \tilde{\mathcal{O}}(h^{-(2+1/r_1)})$ rounds of random exploration in Algorithm 2,*

$$\left| \langle X_t, \hat{\beta}_{s,t_0+1}^{(a)} \rangle - \langle X_t, \beta^{(a)} \rangle \right| \leq h/4, \text{ for all } s \leq s^*, a \in [m], \text{ and any } t > t_0,$$

with probability at least $1 - 3T^{-2}$.

Fix any $s \leq s^*$ and $a \in [m]$. Throughout, we write $\tilde{X}_i = \mathbf{L}_{K_s^{1/2}} X_i$ for the transformed covariates under kernel K_s , and let $\lambda_t = \lambda_{s,t}$ for brevity. The empirical operator of interest is

$$\hat{\mathbf{T}}_{s,t}^{(a)} = \frac{1}{t} \sum_{i=1}^t \mathbb{1}\{a_i = a\} \tilde{X}_i \otimes \tilde{X}_i + \lambda_t \mathbf{1}.$$

We first show that on the event of Lemma EC.4, the full empirical operator $\hat{\mathbf{T}}_{s,t}^{(a)}$ dominates the empirical second-moment operator restricted to outside-margin covariates. For any $i > t_0$ and any covariate $X_i \in \mathcal{U}_h^{(a)}$, the definition of $\mathcal{U}_h^{(a)}$ gives

$$\langle X_i, \beta^{(a)} \rangle - \max_{a' \neq a} \langle X_i, \beta^{(a')} \rangle > h.$$

By Lemma EC.4, on the event of probability at least $1 - 3T^{-2}$, the exploration-phase estimates satisfy $|\langle X_i, \widehat{\beta}_{s', t_0+1}^{(a')} \rangle - \langle X_i, \beta^{(a')} \rangle| \leq h/4$ for all $a' \in [m]$ and all $s' \leq s^*$. In particular, for $s' = 1$,

$$\begin{aligned} \langle X_i, \widehat{\beta}_{1, t_0+1}^{(a)} \rangle - \max_{a' \neq a} \langle X_i, \widehat{\beta}_{1, t_0+1}^{(a')} \rangle &\geq \langle X_i, \beta^{(a)} \rangle - h/4 - \max_{a' \neq a} \left(\langle X_i, \beta^{(a')} \rangle + h/4 \right) \\ &= \langle X_i, \beta^{(a)} \rangle - \max_{a' \neq a} \langle X_i, \beta^{(a')} \rangle - h/2 > h/2 > 0. \end{aligned}$$

Therefore, the condition in line 4 of Algorithm 2 is satisfied, and the algorithm selects $a_i = \arg \max_{a' \in [m]} \langle X_i, \widehat{\beta}_{1, t_0+1}^{(a')} \rangle = a$. Therefore, $\mathbb{1}\{a_i = a\} \geq \mathbb{1}\{X_i \in \mathcal{U}_h^{(a)}\}$ for all $i > t_0$ on the event of Lemma EC.4. Since each summand $\mathbb{1}\{a_i = a\} \widetilde{X}_i \otimes \widetilde{X}_i \succeq 0$, we may drop the exploration-phase terms ($i \leq t_0$) and restrict to outside-margin covariates to obtain

$$\widehat{\mathbf{T}}_{s,t}^{(a)} = \frac{1}{t} \sum_{i=1}^t \mathbb{1}\{a_i = a\} \widetilde{X}_i \otimes \widetilde{X}_i + \lambda_t \mathbf{1} \succeq \frac{1}{t} \sum_{i=t_0+1}^t \mathbb{1}\{X_i \in \mathcal{U}_h^{(a)}\} \widetilde{X}_i \otimes \widetilde{X}_i + \lambda_t \mathbf{1}. \quad (\text{EC.21})$$

Define $Y_i = \mathbb{1}\{X_i \in \mathcal{U}_h^{(a)}\} \widetilde{X}_i$ for $i > t_0$. Since $X_i \sim \mathcal{F}$ independently for each i , the random elements $\{Y_i\}_{i=t_0+1}^t$ are i.i.d. with population second moment operator

$$\mathbb{E}[Y_i \otimes Y_i] = \mathbb{E} \left[\mathbb{1}\{X_i \in \mathcal{U}_h^{(a)}\} \widetilde{X}_i \otimes \widetilde{X}_i \right]. \quad (\text{EC.22})$$

Conjugating both sides of Assumption 8 by $\mathbf{L}_{K_s^{1/2}}$ gives

$$\widetilde{\Sigma}_s \preceq c_o \mathbb{E}[Y_i \otimes Y_i]. \quad (\text{EC.23})$$

We apply Lemma EC.9 to the i.i.d. random elements $Y_i = \mathbb{1}\{X_i \in \mathcal{U}_h^{(a)}\} \widetilde{X}_i$, $i = t_0 + 1, \dots, t$. Note that Y_i is $\widetilde{\Sigma}_s$ -sub-Gaussian. We verify the dominance condition (EC.35) with $\Sigma_0 = \mathbb{E}[Y_i \otimes Y_i]$, $\Sigma_{\text{proxy}} = \widetilde{\Sigma}_s$, and $\lambda = \lambda_t$. By (EC.23),

$$\widetilde{\Sigma}_s + \lambda_t \mathbf{1} \preceq c_o (\mathbb{E}[Y_i \otimes Y_i] + \lambda_t \mathbf{1}),$$

and (EC.35) holds with $c_d = c_o$. By Lemma EC.9 applied to the $t - t_0$ samples $\{Y_i\}_{i=t_0+1}^t$,

$$\frac{1}{t - t_0} \sum_{i=t_0+1}^t Y_i \otimes Y_i + \lambda_t \mathbf{1} \succeq \frac{1}{2} (\mathbb{E}[Y_i \otimes Y_i] + \lambda_t \mathbf{1})$$

with probability at least $1 - \delta_0$ whenever $t - t_0 \geq c'(\lambda_t)^{-1/(2r_s)} \log(1/\delta_0)$ for a sufficiently large constant $c' > 0$ depending on c_o . Substituting into (EC.21) and using $(t - t_0)/t \geq 1/2$ for $t \geq 2t_0$,

$$\widehat{\mathbf{T}}_{s,t}^{(a)} \succeq \frac{t - t_0}{t} \cdot \frac{1}{2} (\mathbb{E}[Y_i \otimes Y_i] + \lambda_t \mathbf{1}) + \frac{t_0}{t} \lambda_t \mathbf{1} \succeq \frac{1}{4} (\mathbb{E}[Y_i \otimes Y_i] + \lambda_t \mathbf{1}) \succeq \frac{1}{4c_o} (\widetilde{\Sigma}_s + \lambda_t \mathbf{1}),$$

where the last inequality uses (EC.23). This establishes (EC.18). Since $t - t_0 \geq t/2$ for $t \geq 2t_0$, the sample size condition is satisfied by the choice $\lambda_t = \lambda_{s,t} \asymp (\log(mT)/t)^{2r_s/(2r_s+1)}$.

Finally, setting $\delta_0 = (lmT)^{-1}T^{-2}$ (so that $\log(1/\delta_0) = \mathcal{O}(\log(lmT))$) and taking a union bound over all $s \in [s^*]$, $a \in [m]$, and $t \in \{t_0 + 1, \dots, T\}$, the concentration in Lemma EC.9 fails with probability at most $lmT \cdot (lmT)^{-1}T^{-2} = T^{-2}$. Combining with the failure probability $3T^{-2}$ of Lemma EC.4, (EC.18) holds with probability at least $1 - 4T^{-2}$.

EC.6.2. Proof of Lemma EC.4

Fix any $a \in [m]$ and any model $s \leq s^*$. For notational brevity, write $\lambda = \lambda_{s,t_0}$. Similarly, we write $\tilde{X}_i = \mathbf{L}_{K_s^{1/2}} X_i$ for the transformed covariates under kernel K_s for any $i \in [t_0]$. For any $i \in [t_0]$ in the random exploration phase, $a_i \sim \text{Uniform}([m])$ independently of X_i , so the population operator is

$$\mathbf{T}^{(a)} := \mathbb{E}[\mathbb{1}\{a_i = a\} \tilde{X}_i \otimes \tilde{X}_i] = \frac{1}{m} \tilde{\Sigma}_s.$$

Since $\mathbf{T}^{(a)} = \frac{1}{m} \tilde{\Sigma}_s$, the dominance condition (EC.35) holds trivially with $\Sigma_0 = \mathbf{T}^{(a)}$, $\Sigma_{\text{proxy}} = \tilde{\Sigma}_s$, $\lambda = \lambda$, and $c_d = m$, since

$$\tilde{\Sigma}_s + \lambda \mathbf{1} = m \mathbf{T}^{(a)} + \lambda \mathbf{1} \preceq m(\mathbf{T}^{(a)} + \lambda \mathbf{1}).$$

Since the random exploration samples $\{\tilde{X}_i, a_i\}_{i=1}^{t_0}$ are i.i.d., we apply Lemma EC.9 to the random elements $\mathbb{1}\{a_i = a\} \tilde{X}_i$, $i = 1, \dots, t_0$. By Assumption 2, $\mathbb{1}\{a_i = a\} \tilde{X}_i$ is $\tilde{\Sigma}_s$ -sub-Gaussian. By Lemma EC.9, when $t_0 \geq c\lambda^{-1/(2r_s)} \log(lmT)$, we obtain

$$\hat{\mathbf{T}}_\lambda^{(a)} \succeq \frac{1}{2}(\mathbf{T}^{(a)} + \lambda \mathbf{1}) = \frac{1}{2m} \tilde{\Sigma}_s + \frac{1}{2} \lambda \mathbf{1} \succeq \frac{1}{2m}(\tilde{\Sigma}_s + \lambda \mathbf{1}) \quad (\text{EC.24})$$

with probability at least $1 - (mlT)^{-1}T^{-2}$.

On the event (EC.24), the condition of Lemma 3 is satisfied with $n = t_0$ and $\delta = (mlT)^{-1}T^{-2}$. Therefore, for all $X \in \mathcal{X}$,

$$\left| \langle X, \hat{\beta}_{s,t_0+1}^{(a)} \rangle - \langle X, \beta^{(a)} \rangle \right| \leq \Delta_{s,t_0}^{(a)}(X) \asymp \sqrt{\lambda \log(mT)} + \sqrt{\frac{\log(mT)}{t_0} \langle \tilde{X}, (\hat{\mathbf{T}}_\lambda^{(a)})^{-1} \tilde{X} \rangle} \quad (\text{EC.25})$$

with probability at least $1 - (mlT)^{-1}T^{-2}$. Fix any $t > t_0$. It remains to show that $\Delta_{s,t_0}^{(a)}(X_t) \leq h/4$ with high probability over $X_t \sim \mathcal{F}$.

By (EC.24), $(\hat{\mathbf{T}}_\lambda^{(a)})^{-1} \preceq 2m(\tilde{\Sigma}_s + \lambda \mathbf{1})^{-1}$, so

$$\langle \tilde{X}_t, (\hat{\mathbf{T}}_\lambda^{(a)})^{-1} \tilde{X}_t \rangle \leq 2m \langle \tilde{X}_t, (\tilde{\Sigma}_s + \lambda \mathbf{1})^{-1} \tilde{X}_t \rangle.$$

Since X_t is independent of the exploration data and $\tilde{X}_t$ is $\tilde{\Sigma}_s$ -sub-Gaussian, by a similar argument as in Lemma EC.1,

$$\langle \tilde{X}_t, (\tilde{\Sigma}_s + \lambda \mathbf{1})^{-1} \tilde{X}_t \rangle \leq 2 \operatorname{Tr}(\tilde{\Sigma}_s (\tilde{\Sigma}_s + \lambda \mathbf{1})^{-1}) + \log(mT) \quad (\text{EC.26})$$

with probability at least $1 - (mlT)^{-1}T^{-2}$. Substituting into (EC.25),

$$\Delta_{s,t_0}^{(a)}(X_t) \lesssim \sqrt{\lambda \log(mT)} + \sqrt{\frac{m \log(mT) (\operatorname{Tr}(\tilde{\Sigma}_s (\tilde{\Sigma}_s + \lambda \mathbf{1})^{-1}) + \log(mT))}{t_0}}$$

with probability at least $1 - 3(mlT)^{-1}T^{-2}$, where we take a union bound over the empirical dominance event (EC.24), the prediction error event (EC.25), and the quadratic form tail (EC.26). It remains to choose t_0 . Substituting $\lambda = \lambda_{s,t_0} \asymp (\log(mT)/t_0)^{2r_s/(2r_s+1)}$, the bias and variance terms in the above display are both of order $\tilde{\mathcal{O}}(t_0^{-r_s/(2r_s+1)})$. Indeed, the bias term satisfies

$$\sqrt{\lambda \log(mT)} \asymp \left(\frac{\log(mT)}{t_0} \right)^{r_s/(2r_s+1)} \cdot \sqrt{\log(mT)} = \tilde{\mathcal{O}}\left(t_0^{-r_s/(2r_s+1)}\right),$$

and by Lemma EC.10, $\operatorname{Tr}(\tilde{\Sigma}_s (\tilde{\Sigma}_s + \lambda \mathbf{1})^{-1}) = \mathcal{O}(\lambda^{-1/(2r_s)}) = \mathcal{O}((t_0/\log(mT))^{1/(2r_s+1)})$, so the variance term satisfies

$$\sqrt{\frac{m \log(mT) (\operatorname{Tr}(\tilde{\Sigma}_s (\tilde{\Sigma}_s + \lambda \mathbf{1})^{-1}) + \log(mT))}{t_0}} = \tilde{\mathcal{O}}\left(t_0^{-r_s/(2r_s+1)}\right).$$

For the bound $\Delta_{s,t_0}^{(a)}(X_t) \leq h/4$ to hold, it suffices that $t_0^{-r_s/(2r_s+1)} \lesssim h$, i.e., $t_0 \geq \tilde{\Omega}(h^{-(2+1/r_s)})$. Since this must hold for all $s \leq s^*$, the most conservative model $s = 1$ (with the smallest decay rate r_1) is binding, giving

$$t_0 \geq \tilde{\Omega}(h^{-(2+1/r_1)}).$$

Taking a union bound over all $a \in [m]$, $s \in [l]$, and $t \in \{t_0 + 1, \dots, T\}$, the total failure probability is at most $mlT \cdot 3(mlT)^{-1}T^{-2} = 3T^{-2}$. This completes the proof.

EC.6.3. Proof of Lemma EC.3

The proof follows the same bias-variance decomposition as Lemma 3, with the empirical operator dominance from Lemma EC.2 playing the role of Lemma 2.

Fix any $s \leq s^*$, $a \in [m]$, and $t > t_0$. Write $\tilde{X}_i = \mathbf{L}_{K_s^{1/2}} X_i$ for any $i \in [t]$ and $\lambda_t = \lambda_{s,t}$. Let $\gamma^{(a)} = (\mathbf{L}_{K_s^{1/2}})^{-1} \beta^{(a)} \in \mathcal{L}^2[0, 1]$ and $\hat{\gamma}^{(a)} = (\mathbf{L}_{K_s^{1/2}})^{-1} \hat{\beta}_{s,t}^{(a)} \in \mathcal{L}^2[0, 1]$. By (5) with $K = K_s$, $\lambda_n = \lambda_t$, and $n = t$,

$$\hat{\gamma}^{(a)} = \left(\hat{\mathbf{T}}_{s,t}^{(a)} \right)^{-1} \left(\hat{\mathbf{T}}_{s,t}^{(a),0} \gamma^{(a)} + g_t^{(a)} \right),$$

where $\widehat{\mathbf{T}}_{s,t}^{(a),0} = \frac{1}{t} \sum_{i=1}^t \mathbb{1}\{a_i = a\} \widetilde{X}_i \otimes \widetilde{X}_i$ and $g_t^{(a)} = \frac{1}{t} \sum_{i=1}^t \varepsilon_i \mathbb{1}\{a_i = a\} \widetilde{X}_i$. The prediction error decomposes as in (EC.6):

$$\left| \langle X_t, \widehat{\beta}_{s,t}^{(a)} \rangle - \langle X_t, \beta^{(a)} \rangle \right| = \left| \langle \widetilde{X}_t, \widehat{\gamma}^{(a)} - \gamma^{(a)} \rangle \right| \leq \underbrace{\left| \langle \widetilde{X}_t, (\widehat{\mathbf{T}}_{s,t}^{(a)})^{-1} \widehat{\mathbf{T}}_{s,t}^{(a),0} \gamma^{(a)} - \gamma^{(a)} \rangle \right|}_{|b_t|} + \underbrace{\left| \langle \widetilde{X}_t, (\widehat{\mathbf{T}}_{s,t}^{(a)})^{-1} g_t^{(a)} \rangle \right|}_{|v_t|}.$$

The bias and variance terms are bounded by the same argument as in the proof of Lemma 3. Since (EC.18) holds by assumption, $\widetilde{\Sigma}_s \preceq c_o(\widetilde{\Sigma}_s + \lambda_t \mathbf{1}) \preceq 4c_o^2 \widehat{\mathbf{T}}_{s,t}^{(a)}$, and the bias analysis (cf. (EC.7)) yields

$$|b_t| \leq \sqrt{8c_o^2 \lambda_t \log(1/\delta)}$$

with probability at least $1 - \delta$, while the variance analysis (cf. (EC.8)) yields

$$|v_t| \leq \sigma_\varepsilon \sqrt{\frac{2 \log(1/\delta)}{t} \langle \widetilde{X}_t, (\widehat{\mathbf{T}}_{s,t}^{(a)})^{-1} \widetilde{X}_t \rangle}$$

with probability at least $1 - \delta$. Combining via a union bound over the bias and variance events,

$$\left| \langle X_t, \widehat{\beta}_{s,t}^{(a)} \rangle - \langle X_t, \beta^{(a)} \rangle \right| \leq \Delta_{s,t}^{(a)}(X_t) \asymp \sqrt{\lambda_t \log(mT)} + \sqrt{\frac{\log(mT)}{t} \langle \mathbf{L}_{K_s^{1/2}} X_t, (\widehat{\mathbf{T}}_{s,t}^{(a)})^{-1} \mathbf{L}_{K_s^{1/2}} X_t \rangle}$$

for all $s \leq s^*$, $a \in [m]$, and $t > t_0$, with probability at least $1 - 3T^{-2}$ (after setting $\delta = l^{-1}m^{-1}$ and taking a union bound over s, a, t , and absorbing constants). This completes the proof.

Appendix

EC.7. Technical lemmas

LEMMA EC.5. *Let w be a σ^2 -subgaussian random variable. Then for any $\delta \in (0, 1)$, we have*

$$\mathbb{P} \left[|w| \geq \sigma \sqrt{2 \log(2/\delta)} \right] \leq \delta.$$

Proof. See Lemma 2.2 in Wainwright (2019). $\square$

LEMMA EC.6. *Suppose that $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. σ^2 -subgaussian random variables. Then for any $a_1, a_2, \dots, a_n \in \mathbb{R}$ and any $\delta \in (0, 1)$, we have*

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n a_i \varepsilon_i \right| \geq \frac{\sigma \sqrt{2 \log(2/\delta) \sum_{i=1}^n a_i^2}}{n} \right] \leq \delta.$$

Proof. Since ε_i are i.i.d. σ^2 -subgaussian, $\frac{1}{n} \sum_{i=1}^n a_i \varepsilon_i$ is $\frac{\sigma^2 \sum_{i=1}^n a_i^2}{n^2}$ -subgaussian by Proposition 2.6.1 in Wainwright (2019). The result then follows from Lemma EC.5. $\square$

LEMMA EC.7 (Sub-Gaussian fourth moment bound). *Let X be a Σ_{proxy} -sub-Gaussian random element in a separable Hilbert space $\mathcal{H}$. Then*

$$\mathbb{E}[\|X\|^2 X \otimes X] \preceq 3 \operatorname{Tr}(\Sigma_{\text{proxy}}) \Sigma_{\text{proxy}}.$$

Proof. For any unit vector $u \in \mathcal{H}$, since $\langle u, X \rangle$ is $\langle u, \Sigma_{\text{proxy}} u \rangle$ -sub-Gaussian, the fourth moment satisfies $\mathbb{E}[\langle u, X \rangle^4] \leq 3 \langle u, \Sigma_{\text{proxy}} u \rangle^2$. Let $\{e_k\}$ be an orthonormal basis of $\mathcal{H}$. By Cauchy-Schwarz,

$$\begin{aligned} \langle u, \mathbb{E}[\|X\|^2 X \otimes X] u \rangle &= \sum_k \mathbb{E}[\langle e_k, X \rangle^2 \langle u, X \rangle^2] \\ &\leq \sum_k \sqrt{\mathbb{E}[\langle e_k, X \rangle^4]} \sqrt{\mathbb{E}[\langle u, X \rangle^4]} \\ &\leq 3 \sum_k \langle e_k, \Sigma_{\text{proxy}} e_k \rangle \langle u, \Sigma_{\text{proxy}} u \rangle = 3 \operatorname{Tr}(\Sigma_{\text{proxy}}) \langle u, \Sigma_{\text{proxy}} u \rangle. \end{aligned}$$

Since u is arbitrary, the result follows. $\square$

LEMMA EC.8 (Covariance operator concentration). *Let $\mathcal{H}$ be a separable Hilbert space and let $X_1, \dots, X_n$ be i.i.d. copies of a Σ_{proxy} -sub-Gaussian random element $X \in \mathcal{H}$ with covariance operator $\Sigma := \mathbb{E}[X \otimes X]$. Let $\mathbf{W}_i := X_i \otimes X_i - \Sigma$ denote the centered operator. Define the uncentered operator variance*

$$\mathbf{V} := n \mathbb{E}[(X_1 \otimes X_1)^2] = n \mathbb{E}[\|X_1\|^2 X_1 \otimes X_1],$$

the variance parameter $\nu^2 := \|\mathbf{V}\|_{\text{op}}$, the effective rank $\mathfrak{r}(\mathbf{V}) := \operatorname{Tr}(\mathbf{V})/\nu^2$, and the sub-exponential parameter $\ell \asymp \|\Sigma_{\text{proxy}}\|_{\text{op}}$ (with an absolute implied constant). Then for any $t \geq \frac{1}{2}(\ell + \sqrt{\ell^2 + 4\nu^2})$,

$$\mathbb{P}\left(\left\|\sum_{i=1}^n (X_i \otimes X_i - \Sigma)\right\|_{\text{op}} > t\right) \leq 14 \mathfrak{r}(\mathbf{V}) \exp\left(-\frac{t^2/2}{\nu^2 + \ell t}\right). \quad (\text{EC.27})$$

Proof. We prove the result for $\mathcal{H} = \mathbb{R}^d$ and note that the extension to separable Hilbert spaces follows by the finite-dimensional approximation argument of [Minsker \(2017, Section 3.2\)](#), which applies since the intrinsic moment condition (EC.28) below is preserved under projection onto finite-dimensional subspaces.²

Since X is Σ_{proxy} -sub-Gaussian, the product $\langle u, X \rangle \langle v, X \rangle$ is sub-exponential with parameter proportional to $\|\Sigma_{\text{proxy}}\|_{\text{op}}$ for any unit vectors u, v ([Vershynin 2018, Lemma 2.7.7](#)). A standard tensorization argument (see, e.g., [Koltchinskii and Lounici \(2017, Proposition 2.1\)](#)) then shows that the intrinsic moment condition

$$\mathbb{E}\left[\|\mathbf{W}_1\|_{\text{op}}^{k-2} \mathbf{W}_1^2\right] \preceq \frac{k!}{2} \ell^{k-2} \mathbb{E}[\mathbf{W}_1^2] \quad (\text{EC.28})$$

² This uses $\|P_F \mathbf{W}_i P_F\|_{\text{op}} \leq \|\mathbf{W}_i\|_{\text{op}}$ and $(P_F \mathbf{W}_i P_F)^2 \preceq P_F \mathbf{W}_i^2 P_F$ for any finite-dimensional subspace $F \subset \mathcal{H}$.

holds for all integers $k \geq 2$ with $\ell \asymp \|\mathbf{\Sigma}_{\text{proxy}}\|_{\text{op}}$.

Let $\mathbf{S}_n = \sum_{i=1}^n \mathbf{W}_i$ and $\phi(x) = e^x - x - 1$. Fix $0 < \theta < 1/\ell$. We bound the operator moment generating function of each summand. Expanding $\mathbb{E}[e^{\theta \mathbf{W}_i}]$ and using $\mathbb{E}[\mathbf{W}_i] = 0$,

$$\mathbb{E}[e^{\theta \mathbf{W}_i}] = \mathbf{1} + \theta^2 \mathbb{E} \left[\mathbf{W}_i \left(\sum_{j=0}^{\infty} \frac{(\theta \mathbf{W}_i)^j}{(j+2)!} \right) \mathbf{W}_i \right].$$

Since $\mathbf{W}_i^j \preceq \|\mathbf{W}_i\|_{\text{op}}^j \mathbf{1}$ for $j \geq 0$ and conjugation preserves the Loewner order,

$$\mathbb{E}[e^{\theta \mathbf{W}_i}] \preceq \mathbf{1} + \theta^2 \sum_{j=0}^{\infty} \frac{\theta^j}{(j+2)!} \mathbb{E} \left[\|\mathbf{W}_i\|_{\text{op}}^j \mathbf{W}_i^2 \right].$$

Applying (EC.28) with $k = j+2$ and summing the geometric series,

$$\mathbb{E}[e^{\theta \mathbf{W}_i}] \preceq \mathbf{1} + \frac{\theta^2}{2(1-\theta\ell)} \mathbb{E}[\mathbf{W}_i^2] \preceq \exp \left(\frac{\theta^2}{2(1-\theta\ell)} \mathbb{E}[\mathbf{W}_i^2] \right), \quad (\text{EC.29})$$

where the last step uses $\mathbf{1} + \mathbf{A} \preceq e^{\mathbf{A}}$ for $\mathbf{A} \succeq 0$. Taking the operator logarithm,

$$\log \mathbb{E}[e^{\theta \mathbf{W}_i}] \preceq \frac{\theta^2}{2(1-\theta\ell)} \mathbb{E}[\mathbf{W}_i^2]. \quad (\text{EC.30})$$

We next convert the tail probability into a trace bound. Since $\phi(x) = e^x - x - 1 \geq 0$ for all x , the operator $\phi(\theta \mathbf{S}_n) \succeq 0$. By the functional calculus, $\lambda_{\max}(\phi(\theta \mathbf{S}_n)) = \phi(\theta \lambda_{\max}(\mathbf{S}_n))$. Since ϕ is increasing on $(0, \infty)$ and $\theta t > 0$, on the event $\{\lambda_{\max}(\mathbf{S}_n) > t\}$ we have

$$\text{Tr } \phi(\theta \mathbf{S}_n) \geq \lambda_{\max}(\phi(\theta \mathbf{S}_n)) = \phi(\theta \lambda_{\max}(\mathbf{S}_n)) > \phi(\theta t).$$

Markov's inequality applied to the non-negative random variable $\text{Tr } \phi(\theta \mathbf{S}_n)$ then gives

$$\mathbb{P}(\lambda_{\max}(\mathbf{S}_n) > t) \leq \mathbb{P}(\text{Tr } \phi(\theta \mathbf{S}_n) > \phi(\theta t)) \leq \frac{\mathbb{E} \text{Tr } \phi(\theta \mathbf{S}_n)}{\phi(\theta t)}. \quad (\text{EC.31})$$

To control $\mathbb{E} \text{Tr } \phi(\theta \mathbf{S}_n)$, write $\exp(\theta \mathbf{S}_n) = \exp(\theta \mathbf{S}_{n-1} + \log e^{\theta \mathbf{W}_n})$. Lieb's concavity theorem and Jensen's inequality give $\mathbb{E}_{n-1} \text{Tr } \exp(\theta \mathbf{S}_{n-1} + \log e^{\theta \mathbf{W}_n}) \leq \text{Tr } \exp(\theta \mathbf{S}_{n-1} + \log \mathbb{E} e^{\theta \mathbf{W}_n})$. Iterating n times and substituting (EC.30),

$$\mathbb{E} \text{Tr } \phi(\theta \mathbf{S}_n) \leq \text{Tr} \left(\exp \left(\frac{n\theta^2}{2(1-\theta\ell)} \mathbb{E}[\mathbf{W}_1^2] \right) - \mathbf{1} \right).$$

Since $\mathbb{E}[\mathbf{W}_1^2] = \mathbb{E}[(X_1 \otimes X_1)^2] - \mathbf{\Sigma}^2 \preceq \mathbb{E}[(X_1 \otimes X_1)^2]$ (as $\mathbf{\Sigma}^2 \succeq 0$), we have $n \mathbb{E}[\mathbf{W}_1^2] \preceq n \mathbb{E}[(X_1 \otimes X_1)^2] = \mathbf{V}$. The monotonicity of $A \mapsto \text{Tr}(e^A)$ then gives

$$\mathbb{E} \text{Tr } \phi(\theta \mathbf{S}_n) \leq \text{Tr} \left(\exp \left(\frac{\theta^2}{2(1-\theta\ell)} \mathbf{V} \right) - \mathbf{1} \right). \quad (\text{EC.32})$$

It remains to extract the effective rank from the right-hand side of (EC.32). Set $g := g(\theta) := \frac{\theta^2}{2(1-\theta\ell)}$. Expanding $\exp(g\mathbf{V}) - \mathbf{1}$ in a power series and factoring each term as $\mathbf{V}^m = \mathbf{V}^{1/2}\mathbf{V}^{m-1}\mathbf{V}^{1/2}$,

$$\exp(g\mathbf{V}) - \mathbf{1} = \sum_{m=1}^{\infty} \frac{g^m}{m!} \mathbf{V}^m = \mathbf{V}^{1/2} \left(\sum_{m=1}^{\infty} \frac{g^m}{m!} \mathbf{V}^{m-1} \right) \mathbf{V}^{1/2}.$$

Since $\mathbf{V} \preceq \nu^2 \mathbf{1}$, we have $\mathbf{V}^{m-1} \preceq \nu^{2(m-1)} \mathbf{1}$ for $m \geq 1$, so

$$\exp(g\mathbf{V}) - \mathbf{1} \preceq \mathbf{V}^{1/2} \left(\sum_{m=1}^{\infty} \frac{(g\nu^2)^m}{m! \nu^2} \right) \mathbf{V}^{1/2} = \frac{e^{g\nu^2} - 1}{\nu^2} \mathbf{V} \preceq \frac{e^{g\nu^2}}{\nu^2} \mathbf{V}. \quad (\text{EC.33})$$

Taking the trace of (EC.33) we get

$$\text{Tr}(\exp(g\mathbf{V}) - \mathbf{1}) \leq \frac{e^{g\nu^2}}{\nu^2} \text{Tr}(\mathbf{V}) = \mathfrak{r}(\mathbf{V}) e^{g\nu^2}.$$

Substituting into (EC.32) gives $\mathbb{E} \text{Tr} \phi(\theta \mathbf{S}_n) \leq \mathfrak{r}(\mathbf{V}) e^{g\nu^2}$. Combining with (EC.31) and the bound $\frac{e^{\theta t}}{\phi(\theta t)} \leq 1 + \frac{6}{(\theta t)^2}$ from Minsker (2017, inequality (3.8)):

$$\mathbb{P}(\lambda_{\max}(\mathbf{S}_n) > t) \leq \frac{\mathfrak{r}(\mathbf{V}) e^{g\nu^2}}{\phi(\theta t)} \leq \mathfrak{r}(\mathbf{V}) \exp(g\nu^2 - \theta t) \left(1 + \frac{6}{(\theta t)^2} \right). \quad (\text{EC.34})$$

Finally, we optimize over θ . Setting $\theta = \frac{t}{\nu^2 + \ell t} \in (0, 1/\ell)$, we compute $1 - \theta\ell = \frac{\nu^2}{\nu^2 + \ell t}$ and

$$g(\theta)\nu^2 = \frac{t^2}{2(\nu^2 + \ell t)}, \quad g(\theta)\nu^2 - \theta t = -\frac{t^2}{2(\nu^2 + \ell t)}.$$

Moreover, $\theta t = \frac{t^2}{\nu^2 + \ell t}$, and whenever $t^2 \geq \nu^2 + \ell t$ (equivalently, $t \geq \frac{1}{2}(\ell + \sqrt{\ell^2 + 4\nu^2})$), the prefactor satisfies $1 + \frac{6}{(\theta t)^2} \leq 7$. Therefore

$$\mathbb{P}(\lambda_{\max}(\mathbf{S}_n) > t) \leq 7 \mathfrak{r}(\mathbf{V}) \exp\left(-\frac{t^2/2}{\nu^2 + \ell t}\right).$$

Repeating the argument with $\mathbf{W}_i$ replaced by $-\mathbf{W}_i$ (which satisfies the same moment condition) and using $\mathbb{P}(\|\mathbf{S}_n\|_{\text{op}} > t) \leq \mathbb{P}(\lambda_{\max}(\mathbf{S}_n) > t) + \mathbb{P}(\lambda_{\max}(-\mathbf{S}_n) > t)$ yields (EC.27). $\square$

LEMMA EC.9 (Empirical operator dominance). *Let $\mathcal{H}$ be a separable Hilbert space and let $X_1, \dots, X_n$ be i.i.d. copies of a Σ_{proxy} -sub-Gaussian random element $X \in \mathcal{H}$ with population second moment $\Sigma_0 := \mathbb{E}[X \otimes X]$. Suppose that*

$$\Sigma_{\text{proxy}} + \lambda \mathbf{1} \preceq c_d(\Sigma_0 + \lambda \mathbf{1}) \quad (\text{EC.35})$$

for some constants $c_d > 0$ and $\lambda > 0$. Then

$$\frac{1}{n} \sum_{i=1}^n X_i \otimes X_i + \lambda \mathbf{1} \succeq \frac{1}{2}(\Sigma_0 + \lambda \mathbf{1}) \quad (\text{EC.36})$$

with probability at least $1 - \delta$ whenever $n \geq c' d_{\text{eff}} \log(d_{\text{eff}}/\delta)$, where $d_{\text{eff}} = \text{Tr}(\Sigma_{\text{proxy}}(\Sigma_0 + \lambda \mathbf{1})^{-1}) + 1$ and $c' > 0$ is a constant depending only on c_d .

Proof. Define the regularized population operator

$$\Sigma_\lambda = \Sigma_0 + \lambda \mathbf{1},$$

the whitened variables $Z_i = \Sigma_\lambda^{-1/2} X_i$, and the centered operators $\mathbf{W}_i = Z_i \otimes Z_i - \mathbb{E}[Z_i \otimes Z_i]$.

Then $\mathbb{E}[Z_i \otimes Z_i] = \Sigma_\lambda^{-1/2} \Sigma_0 \Sigma_\lambda^{-1/2} = \mathbf{1} - \lambda \Sigma_\lambda^{-1}$, and

$$\frac{1}{n} \sum_{i=1}^n Z_i \otimes Z_i = \Sigma_\lambda^{-1/2} \left(\frac{1}{n} \sum_{i=1}^n X_i \otimes X_i \right) \Sigma_\lambda^{-1/2}.$$

If $\left\| \frac{1}{n} \sum_{i=1}^n \mathbf{W}_i \right\|_{\text{op}} \leq \frac{1}{2}$, then since $\mathbb{E}[Z_i \otimes Z_i] \preceq \mathbf{1}$,

$$\frac{1}{n} \sum_{i=1}^n Z_i \otimes Z_i = \mathbb{E}[Z_i \otimes Z_i] + \frac{1}{n} \sum_{i=1}^n \mathbf{W}_i \succeq \mathbb{E}[Z_i \otimes Z_i] - \frac{1}{2} \mathbf{1} \succeq \frac{1}{2} \mathbb{E}[Z_i \otimes Z_i].$$

Conjugating by $\Sigma_\lambda^{1/2}$ gives $\frac{1}{n} \sum_{i=1}^n X_i \otimes X_i \succeq \frac{1}{2} \Sigma_0$, and adding $\lambda \mathbf{1}$ to both sides,

$$\frac{1}{n} \sum_{i=1}^n X_i \otimes X_i + \lambda \mathbf{1} \succeq \frac{1}{2} \Sigma_0 + \lambda \mathbf{1} = \frac{1}{2} \Sigma_\lambda + \frac{1}{2} \lambda \mathbf{1} \succeq \frac{1}{2} \Sigma_\lambda,$$

establishing (EC.36).

It remains to show $\left\| \frac{1}{n} \sum \mathbf{W}_i \right\|_{\text{op}} \leq \frac{1}{2}$ with the stated probability. Since X is Σ_{proxy} -sub-Gaussian, Z_i is $\Sigma_\lambda^{-1/2} \Sigma_{\text{proxy}} \Sigma_\lambda^{-1/2}$ -sub-Gaussian. By (EC.35),

$$\Sigma_{\text{proxy}} \preceq c_d(\Sigma_0 + \lambda \mathbf{1}) - \lambda \mathbf{1} \preceq c_d \Sigma_\lambda,$$

so $\|\Sigma_\lambda^{-1/2} \Sigma_{\text{proxy}} \Sigma_\lambda^{-1/2}\|_{\text{op}} \leq c_d$, so Lemma EC.8 gives the sub-exponential parameter $\ell \asymp c_d$. By Lemma EC.7, the uncentered operator variance $\mathbf{V} = n \mathbb{E}[\|Z_i\|^2 Z_i \otimes Z_i]$ satisfies

$$\nu^2 := \|\mathbf{V}\|_{\text{op}} \leq 3nc_d \text{Tr}(\Sigma_{\text{proxy}} \Sigma_\lambda^{-1}),$$

and via the same effective rank calculation as in the proof of Lemma 2 (cf. (EC.4)),

$$\mathfrak{r}(\mathbf{V}) = \frac{\text{Tr}(\mathbf{V})}{\nu^2} = \mathcal{O}(\text{Tr}^2(\Sigma_{\text{proxy}} \Sigma_\lambda^{-1})).$$

Applying Lemma EC.8 with threshold $t = n/2$, the failure probability is at most

$$14\mathfrak{r}(\mathbf{V}) \exp\left(-\frac{n^2/8}{\nu^2 + \ell n/2}\right).$$

For $n \geq c' d_{\text{eff}} \log(d_{\text{eff}}/\delta)$ with $d_{\text{eff}} = \text{Tr}(\mathbf{\Sigma}_{\text{proxy}} \mathbf{\Sigma}_{\lambda}^{-1}) + 1$ and a sufficiently large constant $c' > 0$ depending on c_d , we have $n^2/8 \geq (\nu^2 + \ell n/2) \log(14\mathfrak{r}(\mathbf{V})/\delta)$, and the right-hand side is at most δ . $\square$

LEMMA EC.10 (**Effective dimension bound**). *Under Assumption 3 with $2r > 1$, for any $\lambda > 0$,*

$$\text{Tr}\left(\tilde{\mathbf{\Sigma}}(\tilde{\mathbf{\Sigma}} + \lambda \mathbf{1})^{-1}\right) = \mathcal{O}(\lambda^{-1/(2r)}).$$

Proof. Write $k^* = \lfloor \lambda^{-1/(2r)} \rfloor$. If $k^* \geq 1$, then $\sigma_{k^*} = \mathcal{O}(k^{*-2r}) = \mathcal{O}(\lambda)$. Splitting the sum at k^* ,

$$\text{Tr}\left(\tilde{\mathbf{\Sigma}}(\tilde{\mathbf{\Sigma}} + \lambda \mathbf{1})^{-1}\right) = \sum_{k=1}^{\infty} \frac{\sigma_k}{\sigma_k + \lambda} \leq \sum_{k \leq k^*} 1 + \frac{1}{\lambda} \sum_{k > k^*} \sigma_k \leq k^* + \frac{C}{\lambda} \int_{k^*}^{\infty} x^{-2r} dx = k^* + \frac{C(k^*)^{1-2r}}{(2r-1)\lambda},$$

where the integral converges since $2r > 1$, and $C > 0$ is the constant from $\sigma_k \leq Ck^{-2r}$. Since $(k^*)^{1-2r}/\lambda = \mathcal{O}(\lambda^{(2r-1)/(2r)}/\lambda) = \mathcal{O}(\lambda^{-1/(2r)})$ and $k^* = \mathcal{O}(\lambda^{-1/(2r)})$, both terms are $\mathcal{O}(\lambda^{-1/(2r)})$. If $k^* = 0$ (i.e., $\lambda > 1$), the first sum is empty and

$$\sum_{k=1}^{\infty} \frac{\sigma_k}{\sigma_k + \lambda} \leq \frac{1}{\lambda} \sum_{k=1}^{\infty} \sigma_k = \frac{\text{Tr}(\tilde{\mathbf{\Sigma}})}{\lambda} = \mathcal{O}(\lambda^{-1}) \leq \mathcal{O}(\lambda^{-1/(2r)}),$$

where $\text{Tr}(\tilde{\mathbf{\Sigma}}) < \infty$ since $2r > 1$, and $\lambda^{-1} \leq \lambda^{-1/(2r)}$ since $\lambda > 1$. $\square$